

生态研究前沿速览_2026-03-21

Ecological & Environmental Sciences Journal Digest

2026 年 3 月 21 日 (周六)

共收录 200 篇最新研究 · 48 篇含中文解读 · 22 张图形摘要 · 106 个 RSS 源

□ 本期期刊分类覆盖:

- **【环境与气候】**: 39 篇
- **【海洋与水生】**: 27 篇
- **【生态学核心】**: 25 篇
- **【植物与农业】**: 25 篇
- **【综合顶刊】**: 22 篇
- **【土壤与生物地球化学】**: 17 篇
- **【保护与生物多样性】**: 17 篇
- **【预印本】**: 15 篇
- **【遥感与模型】**: 13 篇

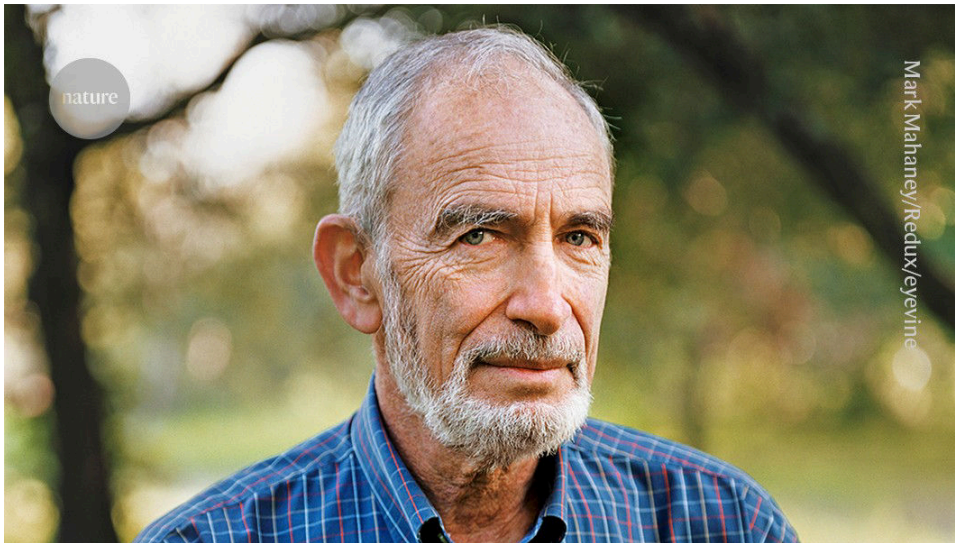
【综合顶刊】

1. Paul R. Ehrlich obituary: pioneering ecologist who caused controversy by predicting a ‘population bomb’

【保罗·R·埃利希讣告：因预言‘人口爆炸’引发争议的先驱生态学家】

□ Nature | 2026-03-20 **【Latest】** | 重要性评分: 40

□ Graphical Abstract:



☐ Abstract:

Nature, Published online: 20 March 2026; doi:10.1038/d41586-026-00939-5 The environmentalist founded the field of co-evolution, and warned that rapid human expansion would cause biodiversity collapse and famine.

☐ 中文摘要:

保罗·埃利希是协同进化领域的创始人，曾警告人类快速扩张将导致生物多样性崩溃和饥荒。他的《人口爆炸》一书虽然预测未完全实现，但深刻影响了环保运动和人口政策讨论。

☐ 深度解读:

Nature 刊登了保罗·埃利希的讣告，纪念这位 20 世纪最具影响力也最具争议性的生态学家。埃利希与 Peter Raven 共同开创的协同进化理论是进化生态学的基石，而他在《人口爆炸》中关于资源枯竭和生态崩溃的警告虽有争议，却催生了现代环保运动和可持续发展理念。这一讣告也引发了关于生态学家公共参与和科学传播角色的当代反思。

☐ DOI: <https://doi.org/10.1038/d41586-026-00939-5>

2. I paused my PhD for 11 years to help save Madagascar's seas

【我暂停博士学位 11 年以拯救马达加斯加的海洋】

☐ Nature | 2026-03-20 【Latest】 | 重要性评分: 38

☐ Graphical Abstract:



☐ Abstract:

Nature, Published online: 20 March 2026; doi:10.1038/d41586-026-00401-6 Marine scientist Ando Rabearisoa creates sustainable, community-managed fisheries.

☐ 中文摘要:

海洋科学家 Ando Rabearisoa 创建了可持续的、社区管理的渔业模式。他暂停博士学业 11 年，投身马达加斯加沿海社区渔业管理的第一线。

☐ 深度解读:

Nature 刊登的这篇个人故事展现了保护科学家在全球南方的真实挑战。Rabearisoa 的经历揭示了学术研究与基层保护实践之间的张力——论文发表的时间线常常与社区保护的紧迫需求不兼容。这一叙事对中国海洋科学工作者参与南海渔业社区共管具有深刻共鸣。

☐ DOI: <https://doi.org/10.1038/d41586-026-00401-6>

3. Chemical pollutants are rife across the world's oceans

【化学污染物在全球海洋中普遍存在】

☐ Nature | 2026-03-20 【Latest】 | 重要性评分: 36

☐ Graphical Abstract:



☐ Abstract:

Nature, Published online: 20 March 2026; doi:10.1038/d41586-026-00850-zCompounds that are used to make plastics and personal-care products were found in all types of marine environment, a meta-analysis shows.

☐ 中文摘要:

用于制造塑料和个人护理产品的化合物在所有类型的海洋环境中均有发现。这项荟萃分析揭示了人为化学品对全球海洋的广泛渗透。

☐ 深度解读:

本研究发表于 Nature，以新闻与评论形式报道了海洋化学污染的全球性规模。配合同期 Nature Geoscience 的研究论文，这一报道将海洋新污染物问题推上了公共议程的优先位置。对中国推进海洋生态文明建设和新污染物治理行动方案具有媒体传播和政策推动价值。

☐ DOI: <https://doi.org/10.1038/d41586-026-00850-z>

4. A wireless sweat sensing with a pH-based correlation model for continuous glucose monitoring and diabetes management during exercise

☐ PNAS | 2026-03-09T07:00:00Z 【Latest】 | 重要性评分: 36

☐ Abstract:

SignificanceAchieving accurate and continuous noninvasive glucose monitoring during exercise is crucial for both diabetic rehabilitation and athletic nutritional management. However, the main challenge of sweat-based glucose monitoring is the weak ...

☐ DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2532127123>

5. Peat fires contribute disproportionately to Siberian fire carbon emissions

【泥炭火灾对西伯利亚火灾碳排放的贡献不成比例】

☐ Science Advances | 2026-03-18T07:00:00Z 【Latest】 | 重要性评分: 34

☐ 中文摘要:

本研究发现泥炭地火灾虽然面积占比相对较小，但对西伯利亚火灾总碳排放的贡献却不成比例地高。泥炭层的深厚有机碳积累使得泥炭火灾的单位面积碳排放远超普通地表火灾。

☐ 深度解读:

本研究发表于 Science Advances，揭示了北方泥炭地火灾在全球碳循环中被严重低估的作用。西伯利亚泥炭地储存了大量碳，泥炭火灾可以燃烧到地下深处，释放数千年积累的碳库。随着气候变暖导致北方火灾频率和强度增加，泥炭火灾的碳排放风险将进一步上升，形成火灾-碳释放-增温的正反馈循环。对全球碳收支核算和中国大兴安岭泥炭地保护管理具有重要参考价值。

☐ DOI: <https://doi.org/10.1126/sciadv.adl2368>

6. From chronic pain to depression: Neurogenesis-driven microglial remodeling in the hippocampal dentate gyrus

☐ Science | 2026-03-19T07:00:00Z 【Latest】 | 重要性评分: 32

☐ DOI: <https://doi.org/10.1126/science.aee6177>

7. Sugarcane streak mosaic virus P1 protein targets the BES1 transcription factor to subvert DAMP-triggered immunity in plant

☐ Science Advances | 2026-03-20T07:00:00Z 【Latest】 | 重要性评分: 32

☐ DOI: <https://doi.org/10.1126/sciadv.adr9432>

8. A disease-agnostic approach to ensemble learning for infectious disease forecasting

☐ Nature Communications | 2026-03-20 【Latest】 | 重要性评分: 32

☐ Abstract:

Nature Communications, Published online: 20 March 2026; doi:10.1038/s41467-026-70937-8 Multi-model ensembles of infectious disease forecasts tend to perform better than individual forecasts, but assigning appropriate weights to component models in the ensemble is challenging. Here, the authors present epiFFORMA, a method for automatic component weighting that does not require historical data.

☐ DOI: <https://doi.org/10.1038/s41467-026-70937-8>

9. One-step fabrication of superhydrophobic fabrics with stable mechanical performance in harsh conditions

☐ Nature Communications | 2026-03-20 【Latest】 | 重要性评分: 32

☐ Abstract:

Nature Communications, Published online: 20 March 2026; doi:10.1038/s41467-026-70857-7 Developing durable, water-repellent textiles without using harmful fluorinated chemicals is challenging. Here, authors use a one-step molecular assembly to covalently bond superhydrophobic silica shells directly onto fibers, resulting in high wear resistance and breathability.

☐ DOI: <https://doi.org/10.1038/s41467-026-70857-7>

10. Copper single-atom nanozyme with intelligent capture and photo-enhanced activity for controlling plant bacterial diseases

☐ Nature Communications | 2026-03-20 【Latest】 | 重要性评分: 32

☐ Abstract:

Nature Communications, Published online: 20 March 2026; doi:10.1038/s41467-026-70930-1 Nanozymes are promising for controlling plant bacterial diseases, but conventional nanozymes suffer from low bacterial affinity, inefficient enzyme-like activity, and thus poor antibacterial efficacy. Here, the authors report a copper single-atom nanozyme that outperforms commercial thiodiazole copper in controlling plant bacterial diseases via intelligent capture and photo-enhanced activity.

☐ DOI: <https://doi.org/10.1038/s41467-026-70930-1>

11. Modulating coordinate site occupancy in high-entropy spinel electrocatalysts

☐ Nature Communications | 2026-03-20 【Latest】 | 重要性评分: 32

☐ Abstract:

Nature Communications, Published online: 20 March 2026; doi:10.1038/s41467-026-70982-3 High entropy spinel oxides host multiple metals in one crystal structure with tunable catalytic properties, but controlling coordination

site occupancy of metal cations is difficult. Here, the authors exploit site preference energy differences to direct metal occupancy, boosting conductivity and oxygen evolution activity.

☐ DOI: <https://doi.org/10.1038/s41467-026-70982-3>

12. The path to room-temperature superconductivity: A programmatic approach

☐ PNAS | 2026-03-09T07:00:00Z 【Latest】 | 重要性评分: 32

☐ Abstract:

Room-temperature superconductivity is arguably the greatest challenge in condensed matter physics, with significant practical and commercial implications if it can be solved. There are no physical laws preventing this from occurring; indeed, ...

☐ DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2520324123>

13. Ambient-pressure 151-K superconductivity in HgBa₂Ca₂Cu₃O_{8+δ} via pressure quench

☐ PNAS | 2026-03-09T07:00:00Z 【Latest】 | 重要性评分: 32

☐ Abstract:

SignificanceThe pressure-quench protocol (PQP) demonstrated here establishes a paradigm for stabilizing at ambient pressure the high-pressure – induced/ – enhanced metastable phases that host elevated superconducting transition temperatures, an effective way ...

☐ DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2536178123>

14. Psoriasis-like disease prevents squamous skin tumor development by neutrophil-driven inflammation

☐ PNAS | 2026-03-09T07:00:00Z 【Latest】 | 重要性评分: 32

☐ Abstract:

SignificanceThis study challenges the prevailing view that chronic inflammation is uniformly tumor-promoting. By demonstrating that psoriasis-like inflammation prevents cutaneous squamous cell carcinoma development in different mouse models, it provides ...

☐ DOI: <https://www.pnas.org/doi/abs/10.1073/pnas.2536378123>

15. Intrinsic over extrinsic: Species identity shapes spatial and interannual Mg/Ca patterns in Arctic marine calcifiers

☐ PLOS ONE | 2026-03-20T14:00:00Z 【Latest】 | 重要性评分: 16

☐ Abstract:

by Małgorzata Krzemińska, Emma Humphreys-Williams, Tomasz Krzykawski, Piotr Kukliński Marine calcifiers incorporate magnesium into their calcium carbonate skeletons through processes influenced by both ambient environmental conditions and species-specific physiological regulation. As a result, their carbonate structures can serve as valuable archives of past and present oceanic conditions, provided that biological controls are explicitly considered. This study investigated how skeletal magnesium-to-calcium (Mg/Ca) ratios vary in space and time among three Arctic benthic invertebrates differing in phylogeny, evolutionary history, and biomineralization strategy: the barnacle *Semibalanus balanoides*, the spirorbid *Paradexiospira violacea*, and the bryozoan *Harmeria scutulata*. Mg/Ca ratios were

☐ DOI: <https://journals.plos.org/plosone/article>

16. Documenting coral spawning in East Africa: first in situ observations from Zanzibar for three reef-building species

☐ PeerJ | 2026-03-20 【Latest】 | 重要性评分: 16

☐ Abstract:

Broadcast coral spawning is a vital reproductive event for many reef-building species and is essential to the resilience of coral reef ecosystems. Understanding spawning dynamics is key to assessing the reproductive health and resilience of these ecosystems. While extensively documented in the Indo-Pacific and Caribbean, coral spawning remains understudied in the Western Indian Ocean (WIO), a region facing rapid environmental change and increasing coral decline. This study presents the first in situ coral spawning records from Zanzibar for three Scleractinian species: *Galaxea astreata*, *Favites pentagona*, and *Platygyra daedalea*. Observations were made during monthly nighttime monitoring dives from November 2024 to February 2025, timed with lunar and solar cycles linked to peak spawning in o

☐ DOI: <https://peerj.com/articles/20999>

17. Hybrid nature–infrastructure adaptation shapes multidecadal mangrove–shoreline dynamics in a tropical delta

☐ Communications Earth & Environment | 2026-03-19 【Latest】 | 重要性评分: 15

☐ Abstract:

Communications Earth & Environment, Published online: 19 March 2026; doi:10.1038/s43247-026-03402-5 Shoreline stability in sediment-limited deltas can depend as much on engineered dykes as on natural sediment supply, according to an analysis that uses 36 years of satellite-derived shoreline, mangrove, and sediment data, combined with spatial modelling

☐ DOI: <https://doi.org/10.1038/s43247-026-03402-5>

18. A new teleosaurid (Crocodylomorpha, Thalattosuchia) from the Sibumasu Terrane of South-east Asia and a taxonomic reassessment of *Indosinosuchus*

☐ PeerJ | 2026-03-20 【Latest】 | 重要性评分: 14

☐ Abstract:

We describe teleosaurid remains from the Middle to Upper Jurassic Khlong Min Formation at Ban Nam Pun in southern Thailand and define *Indosinosuchus peninsularenensis* sp. nov. This new species is diagnosed by a unique combination of cranial and postcranial features, including the nasal reaching anteriorly at the level of the 18th maxillary alveolus, absence of an incisive foramen, oval-shaped external nares, and broad ornamented osteoderms. Phylogenetic analysis positions *I. peninsularenensis* within a polytomy alongside *Mystriosaurus laurillardii* and other Asian teleosaurids, supporting a monophyletic Teleosauroidae. In addition, the revision of *Indosinosuchus kalasinensis* as a junior synonym of *I. potamosiamensis* also strengthens the taxonomic framework of the genus *Indosinosuchus*. *Indosinosuc*

☐ DOI: <https://peerj.com/articles/20944>

19. Decoding bacterial and fungal richness with autoencoders yields a unified ratio indicating soil health and ecological susceptibility

☐ Communications Earth & Environment | 2026-03-19 【Latest】 | 重要性评分: 10

☐ Abstract:

Communications Earth & Environment, Published online: 19 March 2026; doi:10.1038/s43247-026-03398-y The bacterial-to-fungal richness ratio serves as a unified, scalable indicator of microbial balance, soil health and ecological susceptibility, derived from spatial analysis, deep autoencoders, and structural equation modelling of Australian soil microbiomes.

☐ DOI: <https://doi.org/10.1038/s43247-026-03398-y>

20. Lightweight model LMW-YOLO for small object detection in remote sensing images

☐ Scientific Reports | 2026-03-20 **【Latest】** | 重要性评分: 9

☐ Abstract:

Scientific Reports, Published online: 20 March 2026; doi:10.1038/s41598-026-45055-6 Lightweight model LMW-YOLO for small object detection in remote sensing images

☐ DOI: <https://doi.org/10.1038/s41598-026-45055-6>

21. Developmental, regenerative, and behavioral dynamics in acoel reproduction

☐ eLife | Fri, 20 Mar 2026 00: **【Latest】** | 重要性评分: 9

☐ Abstract:

Acoel worms are an enigmatic and understudied animal lineage. Sparse descriptions suggest a diversity of reproductive anatomies across acoels, and likely a corresponding behavioral diversity. Here, we study the reproductive life history of the acoel *Hofstenia miamia*, an emerging lab-tractable model system. We describe *H. miamia*'s reproductive organs, identifying structures previously unknown in acoels. Following worms from zygotes to adulthood, we find that their reproductive organs emerge in a stereotyped sequence as a function of increasing body size. These organs regenerate in a similar sequence after major injuries and are lost in the opposite sequence during starvation-induced de-growth, suggesting that organ growth may be regulated by a single, size-associated program in all contexts

☐ DOI: <https://elifesciences.org/articles/105712>

22. Visual opponent mechanisms and spectral responses in non-primate vertebrates: taxonomic distribution, sampling, and classification

☐ PeerJ | 2026-03-20 **【Latest】** | 重要性评分: 8

☐ Abstract:

Understanding neural mechanisms of opponency is the next frontier of comparative color vision research. Opponent cell properties can be investigated from different perspectives: the type of cone inputs to a cell (cone opponency), and the wavelengths of light to which the cell responds (spectral opponency). Based on a recent database (DOI 10.1016/j.dib.2024.111166) on cone and spectral opponency across non-primate vertebrates, we identified major taxonomic trends in the distribution of opponent cells through image-forming brain regions, analyzed sampling metrics and experimental variables influencing their detection, and established the limitations of existing classification systems. Evidence of opponency is reported in mammals, reptiles (exclusively turtles), amphibians, fish, and birds. W

☐ DOI: <https://doi.org/10.1016/j.dib.2024.111166>

【生态学核心】

23. Responses of African Savanna Trees to Large Herbivore Extinction and Rewilding

【非洲热带稀树草原树木对大型食草动物灭绝与再野化的响应】

□ Ecology Letters | Thu, 12 Mar 2026 11: 【Latest】 | 重要性评分: 53

□ 中文摘要:

本研究检验了非洲热带稀树草原树木对大型食草动物灭绝（历史丧失）和再野化（重新引入）的生态响应，揭示顶级消费者对草原生态系统结构和功能的级联调控效应及其可逆性。

□ 深度解读:

本研究发表于 Ecology Letters，通过比较有大型食草动物（大象、犀牛等）区域的树木种群结构，量化了顶级食草动物对热带稀树草原树木群落的调控效应及再野化后的恢复动态。研究发现食草动物通过改变植被结构（抑制高大乔木、促进灌木扩张）根本性地塑造草原景观，且这种影响在再野化后表现出一定的可逆性但存在时滞。对全球再野化项目的设计和预期效果评估具有重要启示，也为非洲国家公园大型动物恢复项目的科学决策提供依据。

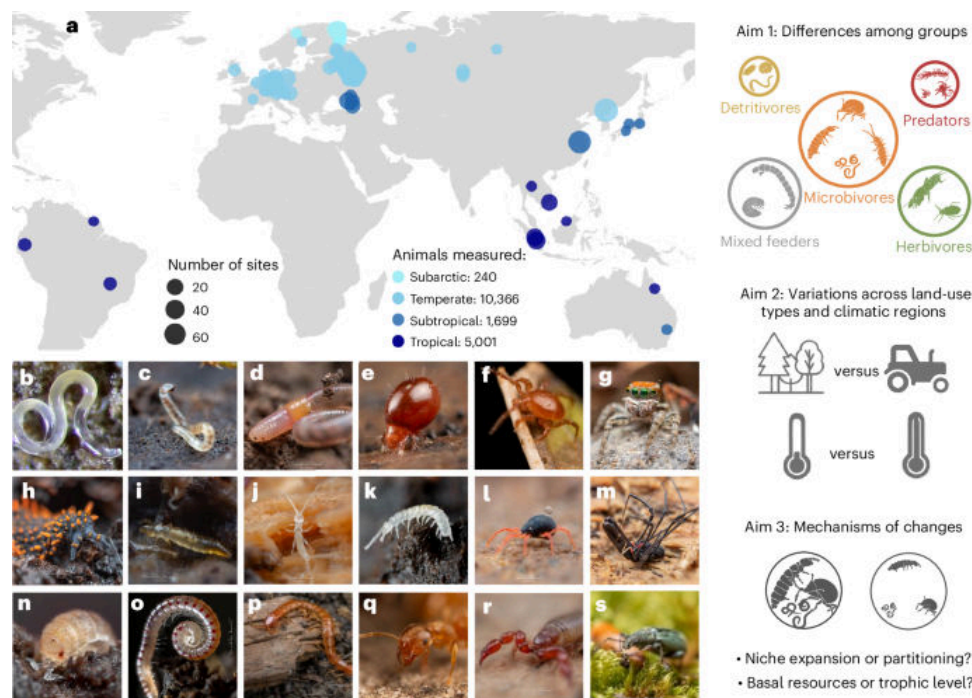
□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70360>

24. Greater trophic diversity of soil animal communities under agricultural land use and tropical climate

【农业用地和热带气候下土壤动物群落具有更高的营养多样性】

□ Nature Ecology & Evolution | 2026-03-16 【Latest】 | 重要性评分: 46

□ Graphical Abstract:



□ Abstract:

Nature Ecology & Evolution, Published online: 16 March 2026; doi:10.1038/s41559-026-03014-4 Soil fauna is an important but often neglected component of terrestrial food webs. Here the authors use a large dataset of stable

isotope observations to analyse how soil animal trophic diversity varies across climates and land-use types and identify potential biotic mechanisms.

□ 中文摘要:

土壤动物是陆地食物网的重要但常被忽视的组分。本研究利用大规模稳定同位素观测数据集，分析土壤动物营养多样性如何随土地利用、气候和土壤性质的变化而变化，揭示农业土地利用和热带气候对土壤动物群落结构的塑造作用。

□ 深度解读:

本研究发表于 Nature Ecology & Evolution，利用全球最大规模的土壤动物稳定同位素数据集，揭示了一个反直觉的发现：农业土地利用虽然通常被认为会降低土壤生物多样性，但实际上可能增加土壤动物的营养功能多样性。这挑战了‘农业简化食物网’的固有认知，表明农业生态系统中的土壤食物网可能比预想的更复杂。对评估农业可持续性和理解土壤生态系统功能提供了新的指标体系。

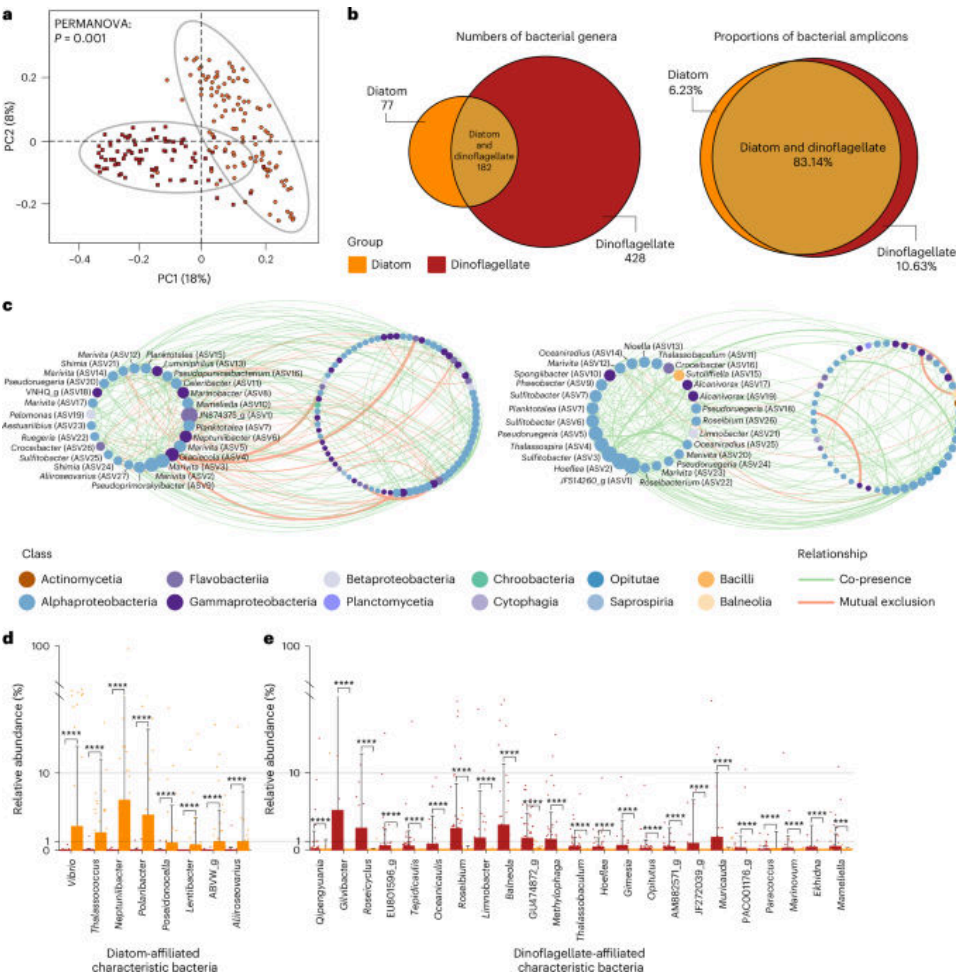
□ DOI: <https://doi.org/10.1038/s41559-026-03014-4>

25. Taxonomic composition and ecophysiology of resident bacteria associated with marine phytoplankton

【与海洋浮游植物共生的常驻细菌的分类组成与生理生态学】

□ Nature Ecology & Evolution | 2026-03-20 【Latest】 | 重要性评分: 43

□ Graphical Abstract:



□ Abstract:

Nature Ecology & Evolution, Published online: 20 March 2026; doi:10.1038/s41559-026-03037-x This study reveals that some bacteria living in the phycosphere of marine diatoms and dinoflagellates are host-specific with genomic functions matching their hosts, while some termed as ‘foundation’ bacteria can span both phycosphere types.

□ 中文摘要:

本研究揭示了与海洋硅藻和甲藻藻际圈（phycosphere）共生的部分细菌具有宿主特异性，其基因组功能与宿主的生理特征相匹配。这些常驻细菌-浮游植物的相互作用对理解海洋微生物食物网和碳循环具有重要意义。

□ 深度解读:

本研究发表于 Nature Ecology & Evolution，首次系统揭示了海洋浮游植物藻际圈常驻细菌的宿主特异性及其生态功能。研究发现这些共生细菌并非随机附着，而是与特定宿主形成了稳定的功能互惠关系——细菌提供维生素和铁载体，浮游植物提供有机碳源。这一发现对理解海洋初级生产力的微生物调控机制具有重要意义，也为预测气候变化下浮游植物群落组成变化的级联效应提供了新视角。

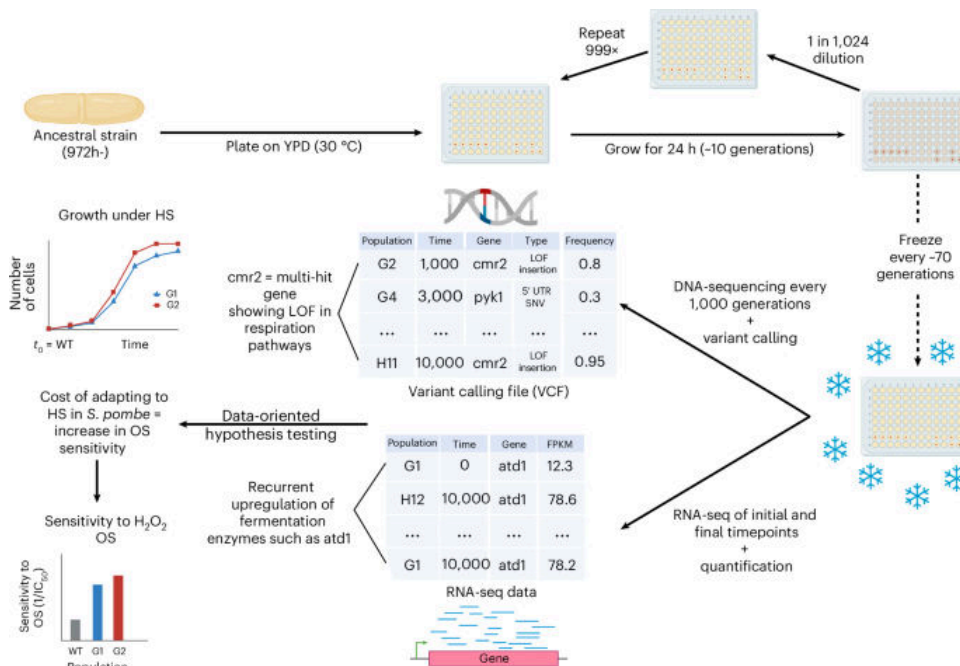
□ DOI: <https://doi.org/10.1038/s41559-026-03037-x>

26. Parallel but distinct adaptive routes in the budding and fission yeasts after 10,000 generations of experimental evolution

【一万代实验进化后出芽酵母和裂殖酵母的平行但截然不同的适应路径】

□ Nature Ecology & Evolution | 2026-03-13 【Latest】 | 重要性评分: 40

□ Graphical Abstract:



□ Abstract:

Nature Ecology & Evolution, Published online: 13 March 2026; doi:10.1038/s41559-026-03017-1 Experimental evolution of fission yeast (*Schizosaccharomyces pombe*) in the same environment as a previous experiment with budding yeast (*Saccharomyces cerevisiae*) reveals parallel evolution but distinct molecular mechanisms and targets of adaptation in the two species.

□ 中文摘要:

在与之前出芽酵母实验相同的环境中进行裂殖酵母实验进化，揭示了平行进化但不同的分子机制。两种远缘酵母在相同选择压力下达到了相似的适应结果，但走了完全不同的遗传路径。

☐ 深度解读:

本研究发表于 *Nature Ecology & Evolution*, 通过比较两种亲缘关系较远的酵母在相同环境下的一万代实验进化, 系统揭示了趋同进化与发散进化共存的规律。研究回答了进化生物学核心问题: 进化是否可重复? 答案是表型水平上可重复, 但分子路径差异巨大。这对理解物种在快速环境变化下的进化响应能力和预测性具有重要意义。

☐ DOI: <https://doi.org/10.1038/s41559-026-03017-1>

27. River Microbiomes as Sentinels of National-Scale Freshwater Ecosystems

【河流微生物组作为全国尺度淡水生态系统的哨兵】

☐ *Global Change Biology* | Wed, 18 Mar 2026 08: **【Latest】** | 重要性评分: 38

☐ 中文摘要:

本研究利用河流微生物组(细菌群落组成)作为全国尺度淡水生态系统健康状况的生物指示器, 揭示了微生物群落结构变化与水质、土地利用和气候等环境因子的关联, 为建立新一代水生态监测体系提供了方法学基础。

☐ 深度解读:

本研究发表于 *Global Change Biology*, 提出了利用河流微生物组进行大尺度水生态系统健康评估的新范式。传统水质监测主要依赖化学指标和大型水生生物, 而微生物群落因其对环境变化的高敏感性和采样便捷性, 可作为更早期的‘生态哨兵’。研究建立的微生物-环境质量关联模型为中国正在推进的全国水生态监测体系升级(从传统理化指标向生物完整性评价转型)提供了重要科学参考。

☐ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70809>

28. Structural characteristics mediate forest mitigation potential against climate change and biodiversity loss

【森林结构特征介导其对气候变化和生物多样性丧失的缓解潜力】

☐ *Ecological Applications* | Mon, 16 Mar 2026 00: **【Latest】** | 重要性评分: 38

☐ 中文摘要:

本研究探讨森林结构特征如何调控森林对气候变化和生物多样性丧失的缓解潜力, 揭示林分结构(胸径多样性、垂直层次、树龄组成)对森林碳汇功能和生物多样性维持的关键调控作用。

☐ 深度解读:

本研究发表于 *Ecological Applications*, 从森林结构功能角度揭示了结构复杂性对森林气候和生物多样性双重缓解功能的调控机制。多层次、多年龄组的结构复杂型森林不仅碳储量更高, 也支撑更丰富的生物多样性。对中国天然林保护工程和人工林近自然化改造具有直接指导意义, 支持推进‘碳汇林’从单一树种人工林转向近自然混交林的政策方向。

☐ DOI: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70211>

29. Genetic ancestry affects cancer immunity

☐ *Nature Ecology & Evolution* | 2026-03-11 **【Latest】** | 重要性评分: 36

☐ Abstract:

Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03012-6 Genetic ancestry affects cancer immunity

☐ DOI: <https://doi.org/10.1038/s41559-026-03012-6>

30. Warming-Induced Carbon Vulnerability in Permafrost Forests: A Shift in Q10 From Continuous to Discontinuous Zones

【多年冻土森林中增温诱导的碳脆弱性：Q10 从连续冻土区到不连续冻土区的转变】

□ Global Change Biology | Wed, 18 Mar 2026 03: **【Latest】** | 重要性评分: 36

□ 中文摘要:

本研究揭示气候变暖导致多年冻土森林碳脆弱性增加的机制，发现土壤呼吸温度敏感性（Q10）从连续冻土区到不连续冻土区存在显著的空间转变，表明冻土退化会改变碳分解的温度响应特性。

□ 深度解读:

本研究发表于 Global Change Biology，揭示了多年冻土退化过程中一个关键的碳循环反馈机制。连续冻土区的土壤碳因长期低温保护而具有较高的 Q10 值（对升温更敏感），一旦冻土开始退化进入不连续冻土阶段，这些‘易感碳’将释放大量 CO₂，形成正反馈。对中国东北大兴安岭和青藏高原冻土区的碳储量安全评估具有重要警示意义。

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70787>

31. A Theoretical Framework for Multispecies Coexistence in Large Herbivores Based on Functional Traits and Dietary Data

【基于功能性状和食性数据的大型食草动物多物种共存理论框架】

□ Ecology Letters | Tue, 17 Mar 2026 21: **【Latest】** | 重要性评分: 36

□ 中文摘要:

本研究提出了一个结合功能性状和食性数据的理论框架，用于解释大型食草动物群落中的多物种共存机制。框架整合了资源划分理论和现代共存理论，为理解草食动物多样性的维持机制提供了新视角。

□ 深度解读:

本研究发表于 Ecology Letters，发展了大型食草动物共存的新理论框架，弥合了经典生态位理论和现代共存理论之间的鸿沟。通过整合功能性状（体型、消化生理、口部形态）和实际食性数据，该框架可以预测不同环境条件下哪些食草动物组合能够稳定共存。对非洲草原、亚洲草原等大型草食动物多样性热点区域的保护管理具有理论指导价值。

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70365>

32. Plasticity in prostate cancer

□ Nature Ecology & Evolution | 2026-03-11 **【Latest】** | 重要性评分: 34

□ Abstract:

Nature Ecology & Evolution, Published online: 11 March 2026; doi:10.1038/s41559-026-03024-2Plasticity in prostate cancer

□ DOI: <https://doi.org/10.1038/s41559-026-03024-2>

33. Earthworms Enhance Global Soil Carbon Storage Through Microbial–Mineral Stabilization

【蚯蚓通过微生物-矿物稳定化增强全球土壤碳储存】

□ Global Change Biology | Thu, 19 Mar 2026 04: **【Latest】** | 重要性评分: 34

□ 中文摘要:

本研究揭示蚯蚓通过促进微生物-矿物有机复合体的形成来增强土壤有机碳的长期稳定储存。蚯蚓的肠道消化和排泄过程改变了有机质与矿物颗粒的结合方式，形成更稳定的碳库。

□ 深度解读:

本研究发表于 Global Change Biology，揭示了蚯蚓这一‘生态系统工程师’在全球碳循环中的关键作用机制。蚯蚓通过消化作用将有机质和矿物颗粒重新组合为更稳定的微团聚体，使碳在土壤中的停留时间延长数十年。这一发现对全球土壤碳固存策略具有重要启示：保护和恢复蚯蚓种群可以作为一种低成本的土壤碳增汇手段，对中国农田土壤碳汇项目和退化土壤修复具有实践指导意义。

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70815>

34. Functional Diversity in Land Surface Modeling: Where and When Does It Matter for the Terrestrial Carbon Cycle?

【陆面模型中的功能多样性：何时何地陆地碳循环至关重要】

□ Global Change Biology | Wed, 18 Mar 2026 03: 【Latest】 | 重要性评分: 34

□ 中文摘要:

本研究探讨在陆面模型中纳入植被功能多样性对陆地碳循环模拟的影响，识别了功能多样性对碳通量预测至关重要的空间区域和时间尺度，为改进地球系统模型中的植被表达方式提供了指导。

□ 深度解读:

本研究发表于 Global Change Biology，系统评估了当前陆面模型中简化植被功能类型（PFT）方法的局限性。发现在生物多样性热点区域（热带雨林、温带混交林）和极端气候事件期间，忽略功能多样性会导致碳通量预测出现显著偏差。这一发现对新一代地球系统模型（如 CMIP7）的植被模块改进具有直接指导意义。

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70785>

35. Top-Down and Bottom-Up Processes Jointly Explain Mesopredator Movement and Foraging Ecology

□ Ecology Letters | Sat, 14 Mar 2026 00: 【Latest】 | 重要性评分: 34

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70364>

36. The Leading Edge Matters Too: Fitness and the Expression of Adaptive Differentiation Are Greatest at the High-Elevation Edge of a Species' Range

□ Ecology Letters | Thu, 12 Mar 2026 11: 【Latest】 | 重要性评分: 34

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70329>

37. Biodiversity Trends Show an Excess of Both Near Stasis and of Very Large Change

【生物多样性趋势显示近乎停滞和极大变化同时过剩】

□ Ecology Letters | Wed, 11 Mar 2026 17: 【Latest】 | 重要性评分: 34

□ 中文摘要:

本研究分析全球生物多样性时间序列数据，发现生物多样性变化趋势呈现两端极值过剩的‘哑铃型’分布：既有大量地点表现出近乎静止的状态，也有大量地点经历了极端变化。

□ 深度解读:

本研究发表于 Ecology Letters，通过分析全球生物多样性数据库（BioTIME 等），揭示生物多样性变化并非简单的‘普遍下降’趋势，而是存在高度异质性——部分生态系统保持稳定，另一部分则经历剧变。这一发现挑战了‘生物多样性普遍下降’的简单化叙事，对生物多样性监测指标设计和 IPBES 评估框架改进具有重要方法论启示。

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/ele.70353>

38. Unprecedented Burning in Tropical Peatlands During the 20th Century Compared to the Previous Two Millennia

□ Global Change Biology | Tue, 17 Mar 2026 03: **【Latest】** | 重要性评分: 30

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70717>

39. Assessing the environmental and dispersal-related drivers of an invasive aquatic plant in Great Lakes coastal wetlands

□ Ecological Applications | Thu, 12 Mar 2026 00: **【Latest】** | 重要性评分: 14

□ DOI: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70206>

40. Effects of temperature and browning on the functional response of a freshwater top predator

□ Journal of Animal Ecology | Mon, 16 Mar 2026 22: **【Latest】** | 重要性评分: 10

□ DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70233>

41. Climate-linked evolution and genetics in a warming Arctic

□ Ecological Monographs | Mon, 16 Mar 2026 05: **【Latest】** | 重要性评分: 10

□ DOI: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70053>

42. Passive environmental DNA sampling: A review of current practices, limitations and future directions for biodiversity monitoring

□ Methods in Ecology and Evolution | Thu, 19 Mar 2026 03: **【Latest】** | 重要性评分: 10

□ DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70247>

43. Environmental filtering drives mycorrhizal tree dominance across a soil fertility gradient in a temperate forest

□ Journal of Ecology | Mon, 16 Mar 2026 03: **【Latest】** | 重要性评分: 8

□ DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70298>

44. Enhanced below-ground functioning is associated with higher plant resistance against drought: Implications for ecosystem functions

□ Journal of Ecology | Fri, 13 Mar 2026 00: **【Latest】** | 重要性评分: 8

□ DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70289>

45. Eco-evolutionary dynamics of partially migratory metapopulations in spatially and seasonally varying environments

☐ Journal of Animal Ecology | Thu, 12 Mar 2026 22: **【Latest】** | 重要性评分: 8

☐ DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70240>

46. Low-nitrogen addition amplifies the positive effect of grazing exclusion on plant diversity and community stability

☐ Functional Ecology | Wed, 18 Mar 2026 07: **【Latest】** | 重要性评分: 8

☐ DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70305>

47. Extending plant water-use strategies to flowers: Evidence from trait correlations across plant organs

☐ Functional Ecology | Tue, 10 Mar 2026 21: **【Latest】** | 重要性评分: 8

☐ DOI: <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70293>

【环境与气候】

48. Sleep loss in a warming world

【变暖世界中的睡眠丧失】

☐ Nature Sustainability | 2026-03-13 **【Latest】** | 重要性评分: 57

☐ Graphical Abstract:



□ Abstract:

Nature Sustainability, Published online: 13 March 2026; doi:10.1038/s41893-026-01778-y High temperatures disrupt sleep worldwide, with disproportionate impacts on older adults, women and populations in lower-income countries. A study uses climate change simulations to project future global sleep erosion and, in turn, the decline in childhood general cognitive ability and associated socioeconomic costs.

□ 中文摘要:

高温在全球范围内扰乱睡眠，对老年人、女性及低收入国家人口的影响尤为严重。本研究利用气候变化模拟预测未来全球睡眠侵蚀的情景，揭示气候变化对人类健康的又一重要但常被忽视的影响路径。

□ 深度解读:

本研究发表于 Nature Sustainability，将气候变化与人类睡眠健康创新性地连接，开拓了气候健康影响研究的新维度。研究利用全球气候模拟数据量化了未来情景下睡眠时长和质量的损失，发现热带和亚热带地区（包括中国南部）面临最大风险。睡眠剥夺将引发认知损害、劳动生产率下降和慢性病发病率上升等级联效应。研究凸显了城市绿化、建筑节能改造等气候适应措施对公共健康的协同效益，为气候政策提供了新的健康代价量化依据。

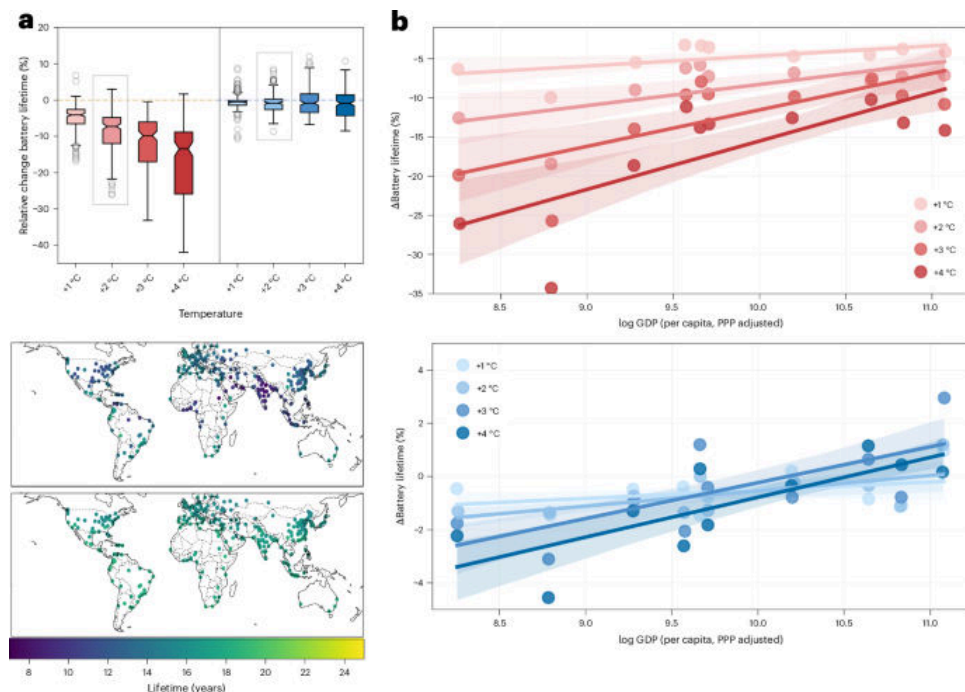
□ DOI: <https://doi.org/10.1038/s41893-026-01778-y>

49. Technological advances mitigate the impact of climate change on electric vehicle battery lifetimes

【技术进步减缓气候变化对电动汽车电池寿命的影响】

□ Nature Climate Change | 2026-03-16 【Latest】 | 重要性评分: 53

□ Graphical Abstract:



□ Abstract:

Nature Climate Change, Published online: 16 March 2026; doi:10.1038/s41558-026-02581-5 We combined electric vehicle simulation and battery degradation models with high-resolution downscaled climate data for 300 global cities. Climate change was predicted to reduce battery lifetime by 8% on average for batteries manufactured between 2010 and 2018 versus 3% for batteries produced after 2019. Thus, technological advances in electric vehicle battery manufacturing demonstrate important climate adaptation co-benefits.

□ 中文摘要:

研究将电动汽车仿真和电池退化模型与 300 个全球城市的高分辨率降尺度气候数据相结合，预测气候变化将平均缩短电池寿命 8%，但技术进步（热管理系统改进、电池化学体系升级）将在主要地区部分或完全抵消这一影响。

□ 深度解读:

本研究发表于 Nature Climate Change，是首个系统量化气候变化对全球电动汽车电池寿命影响及技术对冲潜力的研究。研究利用多城市、多气候情景的大规模模拟，揭示炎热气候（中东、东南亚、中国南部）将是电池寿命受影响最严重的地区。这一发现对中国新能源汽车产业政策具有直接参考价值：既为研发面向热带亚热带气候的电池热管理技术提供了市场导向，也为电动汽车全生命周期碳核算提供了新的气候修正参数。

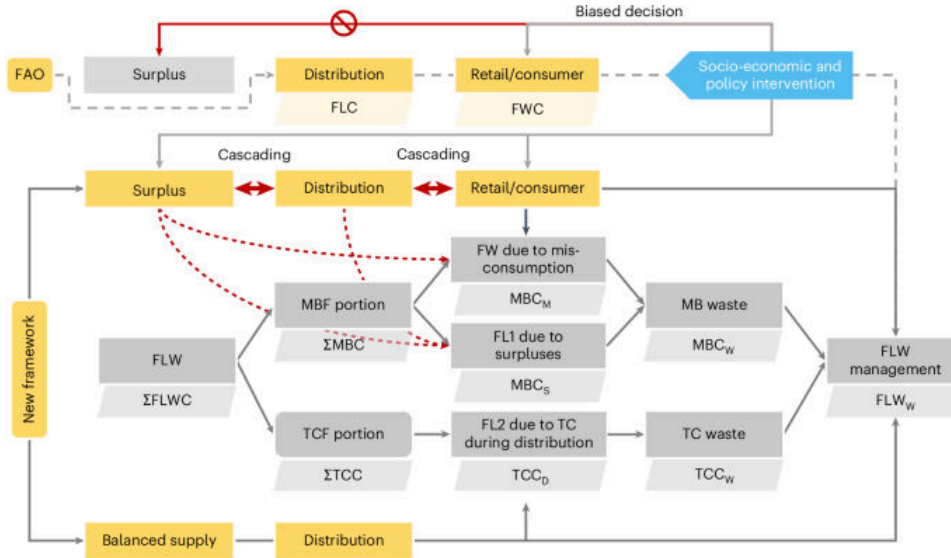
□ DOI: <https://doi.org/10.1038/s41558-026-02581-5>

50. Misbehaviour dominates GHG emissions from food loss and waste

【不当行为主导食品损失和浪费产生的温室气体排放】

□ Nature Climate Change | 2026-03-19 【Latest】 | 重要性评分: 51

□ Graphical Abstract:



□ Abstract:

Nature Climate Change, Published online: 19 March 2026; doi:10.1038/s41558-026-02596-y Food loss and waste (FLW) is a major source of global GHG emissions, yet its drivers and mitigation potential remain understudied. By attributing FLW to techno-economic and misbehavioural drivers, this study shows misbehaviour dominates FLW emissions and offers substantial mitigation potential.

□ 中文摘要:

食品损失和浪费（FLW）是全球温室气体排放的主要来源之一，但其驱动因素和减缓潜力仍研究不足。本研究将 FLW 归因于技术-经济因素和行为因素，发现行为因素（消费者浪费、过度购买等不当行为）主导了 FLW 引起的温室气体排放，这对减排策略的制定具有深刻影响。

□ 深度解读:

本研究发表于 Nature Climate Change，首次系统量化了食品供应链各环节中技术性损失与行为性浪费对温室气体排放的相对贡献。研究发现消费端的‘不当行为’（如超量购买、过早丢弃、储存不当）贡献了食品损耗温室气体排放的大部分，远超供应链中的技术性损失。这一发现将食品减排的政策焦点从供应链效率优化转向消费者行为干预，对中国推进‘反食品浪费法’落实和实现‘双碳’目标中的农食系统减排具有直接政策启示。

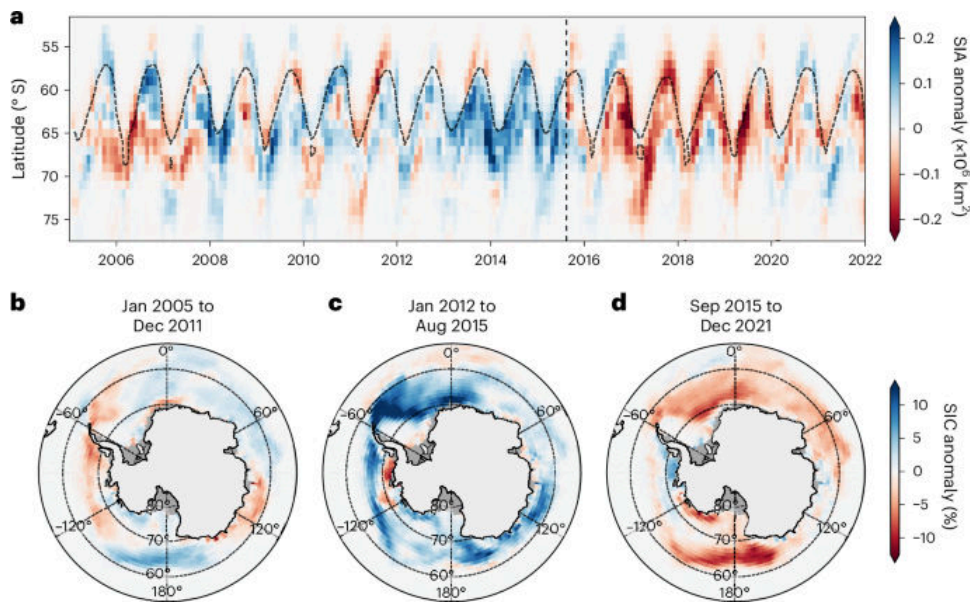
□ DOI: <https://doi.org/10.1038/s41558-026-02596-y>

51. Wind-triggered Antarctic sea-ice decline preconditioned by thinning Winter Water

【冬季水体变薄预设条件下风触发的南极海冰衰退】

□ Nature Climate Change | 2026-03-18 【Latest】 | 重要性评分: 51

□ Graphical Abstract:



□ Abstract:

Nature Climate Change, Published online: 18 March 2026; doi:10.1038/s41558-026-02601-4 Antarctic sea ice declined sharply between 2015 and 2017, and this study uses ocean observations and atmospheric data to determine contributing factors. The authors show that thinning of Winter Water in the previous decade, followed by strong winds, brought warm deep water into contact with sea ice.

□ 中文摘要:

2015 至 2017 年间南极海冰急剧下降，本研究利用海洋观测和大气数据确定了其贡献因素。研究发现冬季水体（一种位于表层混合层下方的冷水层）的变薄为海冰脆弱性创造了前置条件，而异常风场模式则触发了海冰的急剧减少。

□ 深度解读:

本研究发表于 Nature Climate Change，揭示了南极海冰近年急剧衰退的深层海洋学机制。与北极海冰主要受大气增温驱动不同，南极海冰的突变式下降涉及海洋亚表层过程的预置作用——冬季水体的逐渐变薄削弱了海冰下方的冷水‘缓冲层’，使海冰对风场变化更加脆弱。这一‘预置 + 触发’的双阶段机制对理解极地气候突变具有重要理论意义，也提示现有气候模型可能因忽视亚表层海洋过程而低估南极海冰未来的衰退速率。

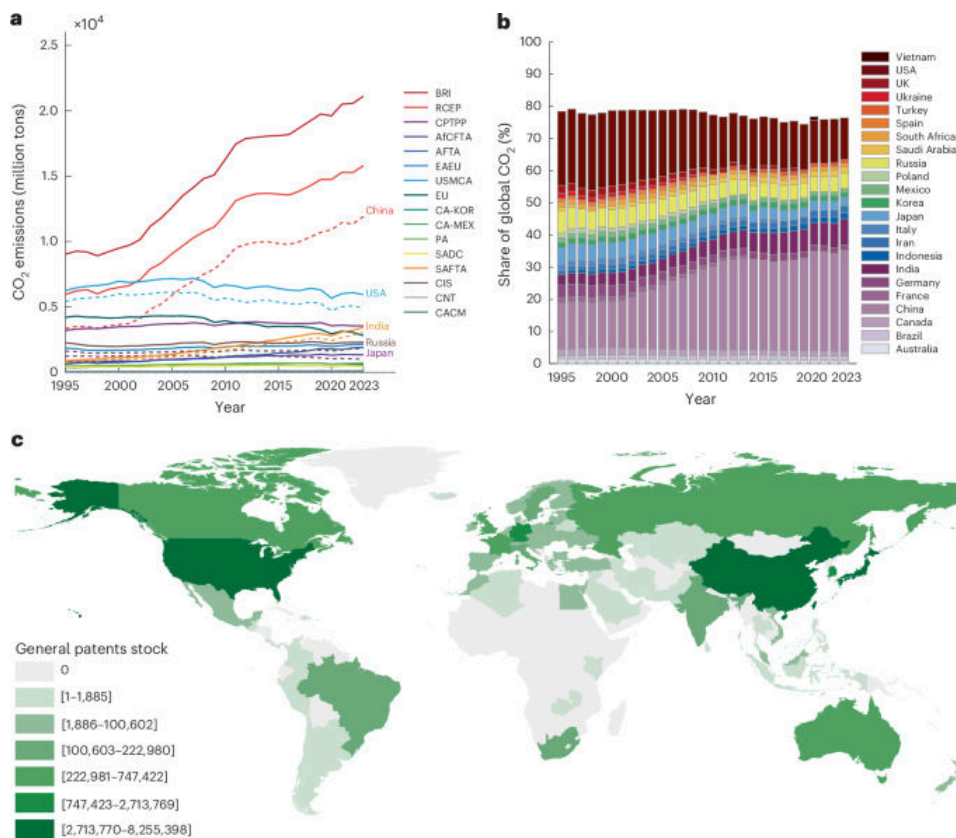
□ DOI: <https://doi.org/10.1038/s41558-026-02601-4>

52. International trade reduces emissions through technology transfer led by key emitters

【国际贸易通过主要排放国主导的技术转让减少排放】

□ Nature Climate Change | 2026-03-17 【Latest】 | 重要性评分: 49

□ Graphical Abstract:



□ Abstract:

Nature Climate Change, Published online: 17 March 2026; doi:10.1038/s41558-026-02595-z Technology advancement is essential for climate action, yet the uneven distribution of technological progress across the world can slow mitigation. Through empirical and scenario analysis, researchers find that participating in trade agreements could enhance technological transfers and lead to emission reductions.

□ 中文摘要:

技术进步对气候行动至关重要，但全球技术进步的不均衡分布可能减缓减缓行动。通过实证和情景分析，研究发现主要排放国（特别是中国）的清洁技术出口通过技术转让机制对全球减排产生了显著溢出效应。

□ 深度解读:

本研究发表于 Nature Climate Change，提供了国际贸易与全球减排关系的实证证据。研究揭示中国等主要排放国通过出口太阳能板、新能源汽车等清洁技术产品，正在成为全球绿色技术扩散的重要驱动力。这一发现对国际气候谈判框架（如贸易与气候政策协调）具有政策含义，也为‘一带一路’绿色发展战略提供了科学支撑。

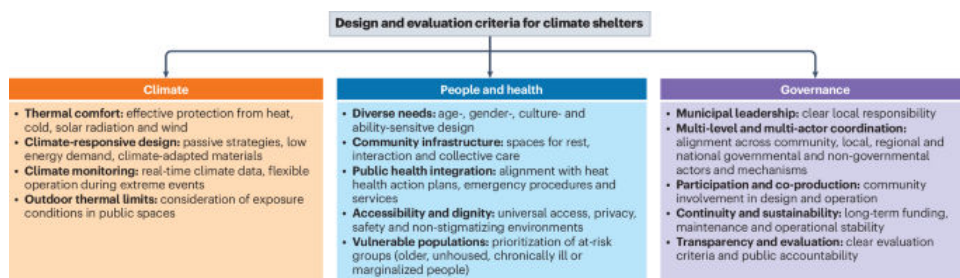
□ DOI: <https://doi.org/10.1038/s41558-026-02595-z>

53. Pioneering Spanish experience in climate shelters practice

【西班牙气候避难所实践的先驱经验】

□ Nature Climate Change | 2026-03-20 【Latest】 | 重要性评分: 47

□ Graphical Abstract:



□ Abstract:

Nature Climate Change, Published online: 20 March 2026; doi:10.1038/s41558-026-02587-zAs cities heat up, climate shelters are increasingly vital for protecting people from extreme heat. Beyond temporary emergency stopgaps, Spain's pioneering experience shows how climate, health and governance align to turn these spaces into enduring infrastructures of care and resilience.

□ 中文摘要:

随着城市升温，气候避难所对保护人们免受极端高温影响日益重要。西班牙的先驱经验不仅仅是临时应急措施，而是将气候避难所整合进城市规划的系统性实践，为全球城市热浪适应策略提供了可复制的模式。

□ 深度解读:

本研究发表于 Nature Climate Change，系统总结了西班牙在城市气候避难所（climate shelters）建设方面的开创性经验。西班牙城市（特别是巴塞罗那）通过将公共图书馆、博物馆、公园等改造为网络化的气候避难所，形成了超越应急措施的系统性城市热适应策略。这一经验对中国城市特别是南方高温频发城市具有直接借鉴意义，对推进韧性城市建设和热浪预警响应体系完善提供了实践参考。

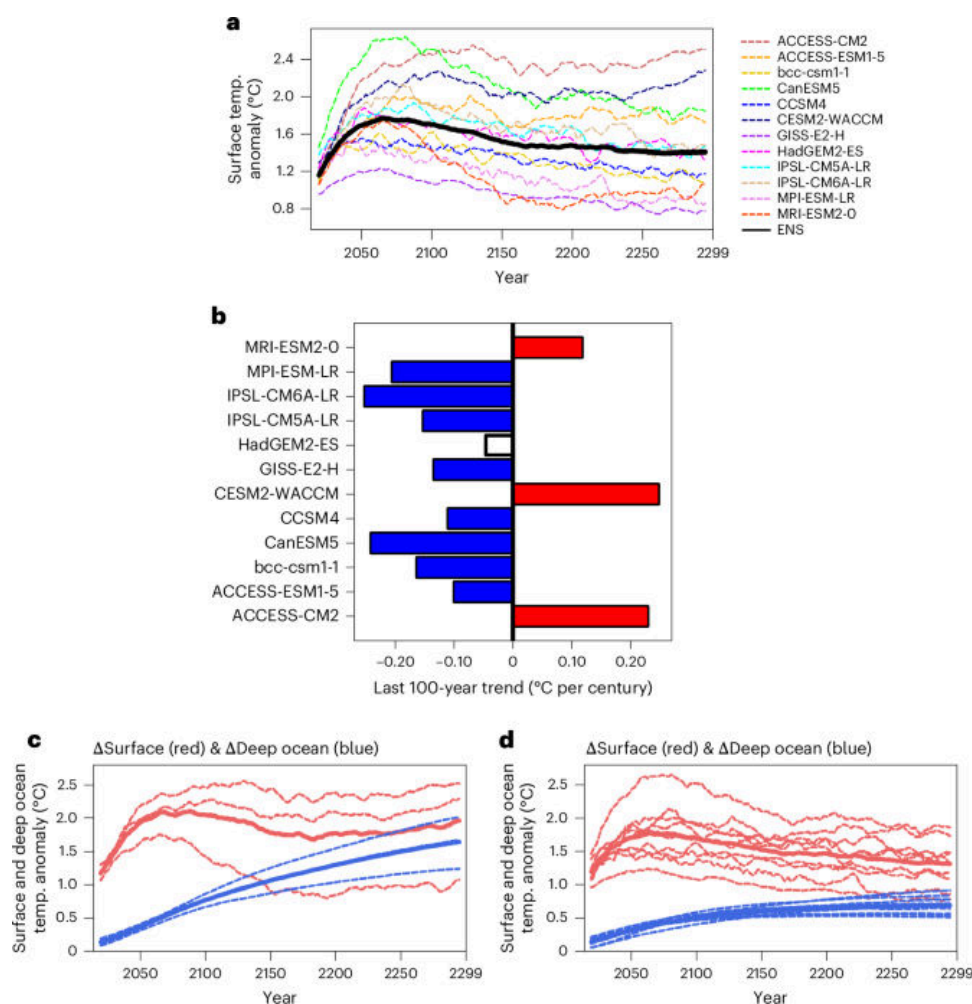
□ DOI: <https://doi.org/10.1038/s41558-026-02587-z>

54. Deep ocean control of global temperature after net-zero emissions

【净零排放后深海对全球温度的调控作用】

□ Nature Geoscience | 2026-03-16 【Latest】 | 重要性评分: 44

□ Graphical Abstract:



□ Abstract:

Nature Geoscience, Published online: 16 March 2026; doi:10.1038/s41561-026-01934-1 When atmospheric carbon dioxide levels decline, increased ocean heat absorption would initially slow surface warming before warming the deep ocean, which ultimately raises global mean surface temperatures, according to future model simulations.

□ 中文摘要:

当大气二氧化碳浓度下降时，海洋吸热量增加将首先减缓地表升温，但随后会加热深海，最终导致全球平均地表温度升高。研究揭示了实现净零排放后的气候演变路径及深海热力学反馈机制。

□ 深度解读:

本研究发表于 Nature Geoscience，揭示了净零排放达成后全球气候演变的一个此前被低估的关键物理机制：深海热容量的延迟释放效应。即使人类实现净零排放，由于深海热容量巨大，地球表面温度仍可能在数十年至数百年内继续上升。这提示实现净零排放并不意味着气候问题立即解决，还需要长期监测和碳移除技术，对全球气候目标理解和政策制定者具有重要警示意义。

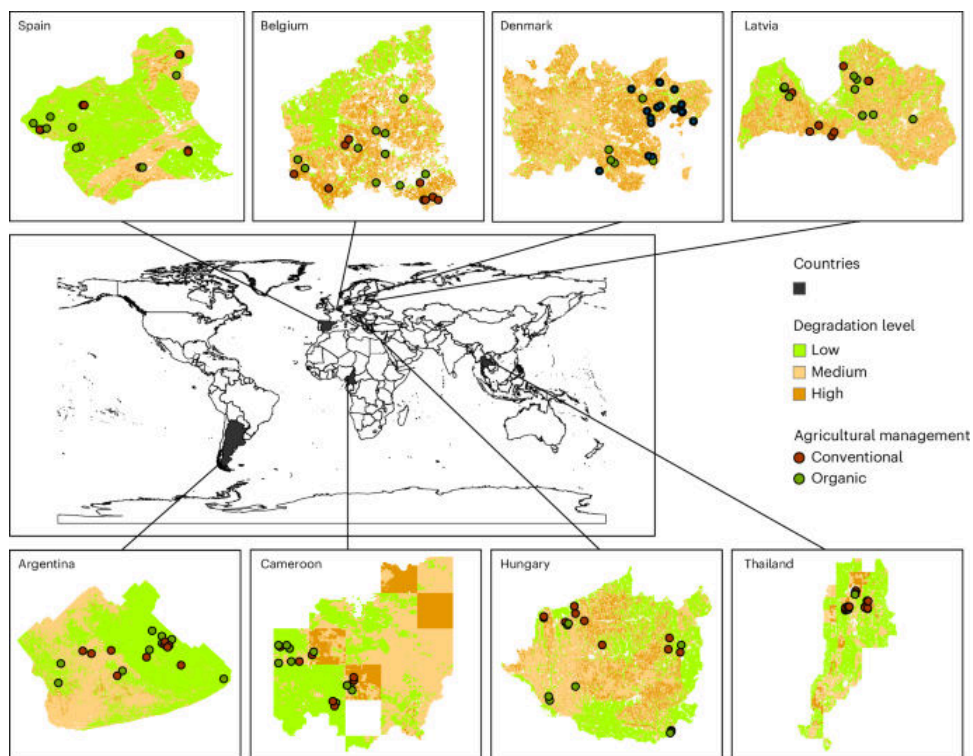
□ DOI: <https://doi.org/10.1038/s41561-026-01934-1>

55. Optimizing biodiversity, multifunctionality and yield when transitioning to organic farming

【向有机农业转型时优化生物多样性、多功能性和产量】

□ Nature Sustainability | 2026-03-16 【Latest】 | 重要性评分: 40

□ Graphical Abstract:



□ Abstract:

Nature Sustainability, Published online: 16 March 2026; doi:10.1038/s41893-026-01791-1 Different agricultural management systems, for example, conventional versus organic, can have different benefits and challenges. Authors here examine the biodiversity, crop yield and ecosystem multifunctionality impacts of transitioning from conventional to organic agriculture across 179 global croplands.

□ 中文摘要:

不同农业管理系统（如常规农业与有机农业）具有不同的效益和挑战。本研究考察了生物多样性、作物产量和生态系统多功能性在不同管理系统中的表现，并探讨了如何在向有机农业转型过程中优化这些目标间的权衡关系。

□ 深度解读:

本研究发表于 Nature Sustainability，通过系统分析欧洲多个长期农业实验数据，揭示了有机农业转型中生物多样性、生态系统多功能性和作物产量之间的协同与权衡关系。发现通过优化农场景观多样性和管理实践组合，可以在产量损失最小的前提下实现生物多样性和生态功能的最大化提升。对中国推进绿色农业发展和昆蒙框架下‘到 2030 年恢复 30% 退化土地’目标的农业用地管理策略选择具有参考意义。

□ DOI: <https://doi.org/10.1038/s41893-026-01791-1>

56. [ASAP] Sulfur Isotopes Reveal Spatial Variation in Waterbird Trace Element Contamination from Tropical Estuaries to the Open Ocean

【硫同位素揭示从热带河口到开阔海洋中水鸟微量元素污染的空间差异】

□ Environmental Science & Technology | Thu, 19 Mar 2026 12: 【Latest】 | 重要性评分: 40

□ Abstract:

Environmental Science & Technology DOI: 10.1021/acs.est.5c14633

□ 中文摘要:

本研究利用硫稳定同位素作为栖息地指示剂，揭示了热带水鸟体内微量元素（重金属等）污染水平从河口到远洋的空间梯度变化。研究建立了将污染源与生物暴露水平关联的新方法框架。

□ 深度解读:

本研究发表于 Environmental Science & Technology, 创新性地利用硫同位素作为水鸟栖息地和食物来源的生物标记物，揭示了不同海洋环境中水鸟面临的差异化污染风险。河口区域水鸟的重金属暴露水平显著高于远洋鸟类，反映了近岸人类活动对海洋食物网的深刻影响。该方法学突破为全球水鸟保护和海洋污染评估提供了新工具。

□ DOI: <https://doi.org/10.1021/acs.est.5c14633>

57. A review of machine learning applications in the prediction of selected groundwater quality parameters: Key lessons, knowledge gaps, and future directions

□ Science of the Total Environment 【Latest】 | 重要性评分: 39

□ Abstract:

Publication date: 25 April 2026 Source: Science of The Total Environment, Volume 1027 Author(s): Ahmed Elsayed, Jana Levison, Andrew Binns, Marie Larocque, Pradeep Goel

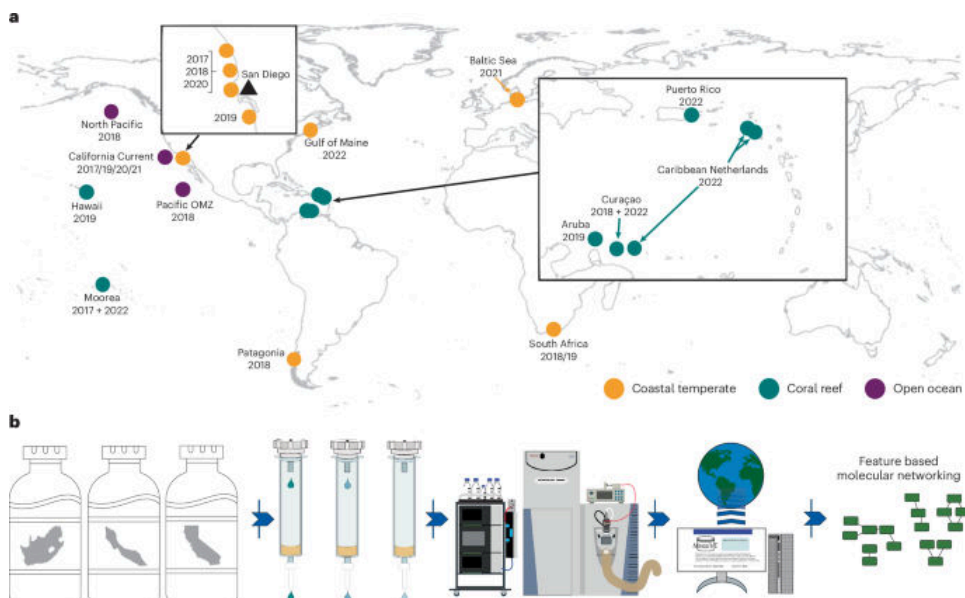
□ DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726003566>

58. Widespread presence of anthropogenic compounds in marine dissolved organic matter

【海洋溶解有机物中人为化合物的广泛存在】

□ Nature Geoscience | 2026-03-16 【Latest】 | 重要性评分: 38

□ Graphical Abstract:



□ Abstract:

Nature Geoscience, Published online: 16 March 2026; doi:10.1038/s41561-026-01928-z Synthetic chemicals can make up substantial proportions of dissolved organic matter in the ocean, with major differences in compound classes depending on the distance from coastlines, according to an analysis of seawater samples from around the globe.

□ 中文摘要:

合成化学品可以占海洋溶解有机物的相当比例，化合物类别的主要差异取决于距海岸线的距离。本研究首次在全球尺度上系统记录了人为合成化学品在海洋溶解有机物库中的广泛分布。

□ 深度解读:

本研究发表于 Nature Geoscience，利用高分辨质谱技术，首次系统揭示了人为合成化学品（持久性有机污染物、药品活性化合物、个人护理品等）在全球海洋溶解有机物中的普遍存在。即使在远离人类活动的深海区域也检测到大量合成化学品，表明海洋已成为全球人为化学品的汇。这对全球新污染物风险评估和中国海洋污染治理具有里程碑意义。

□ DOI: <https://doi.org/10.1038/s41561-026-01928-z>

59. Early action for coastal communities

【对沿海社区采取早期行动】

□ Nature Geoscience | 2026-03-12 【Latest】 | 重要性评分: 36

□ Graphical Abstract:



□ Abstract:

Nature Geoscience, Published online: 12 March 2026; doi:10.1038/s41561-026-01932-3 Shifts in large-scale climate patterns are reshaping flood risk worldwide. Advances in modelling now offer the potential to provide early warnings and develop effective tools for managing rising coastal hazards.

□ 中文摘要:

大尺度气候模式的转变正在重塑全球洪水风险格局。建模技术的进步为提供预警和开发有效应对沿海灾害工具提供了可能。本研究提出针对沿海社区的气候风险早期预警和适应行动框架。

□ 深度解读:

本研究发表于 Nature Geoscience，利用最新一代高分辨率气候模式评估了全球沿海洪涝风险的时空演变格局。特别关注气候模式转变（如 ENSO 频率变化、热带气旋路径移动）对沿海洪水风险的非线性影响。对中国长三角、珠三角等高密度沿海城市群的气候适应规划和国家气候适应战略的制定实施提供了科学支撑。

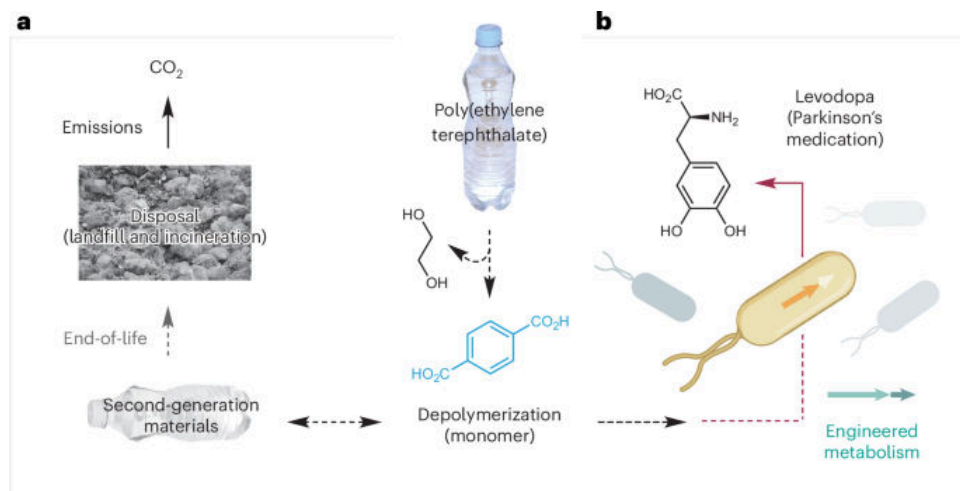
□ DOI: <https://doi.org/10.1038/s41561-026-01932-3>

60. Microbial upcycling of plastic waste to levodopa

【微生物将塑料废物升级利用生产左旋多巴】

□ Nature Sustainability | 2026-03-16 【Latest】 | 重要性评分: 36

□ Graphical Abstract:



□ Abstract:

Nature Sustainability, Published online: 16 March 2026; doi:10.1038/s41893-026-01785-z The production of high-value chemicals can involve energy-intensive processes, necessitating sustainable production strategies. Here the authors present a circular bioeconomy approach, upcycling plastic waste through microbial conversion into levodopa, a medicine for Parkinson's disease.

□ 中文摘要:

高价值化学品的生产往往涉及能源密集型工艺。本研究提出一种循环生物经济方法，通过微生物代谢途径将塑料废物升级转化为左旋多巴（帕金森症的关键药物），实现塑料污染的高值化利用。

□ 深度解读:

本研究发表于 Nature Sustainability，展示了将塑料污染治理与高价值化学品生物制造相结合的创新路径。利用合成生物学工具改造微生物代谢网络，使其以降解塑料产生的小分子为原料合成左旋多巴，实现了‘废物变良药’的范式转变。对中国‘无废城市’建设和合成生物学产业发展具有重要启示。

□ DOI: <https://doi.org/10.1038/s41893-026-01785-z>

61. Iran's policy priorities intensify water crisis

□ Nature Sustainability | 2026-03-13 【Latest】 | 重要性评分: 36

□ Graphical Abstract:



☐ Abstract:

Nature Sustainability, Published online: 13 March 2026; doi:10.1038/s41893-026-01784-0 The chronic water crisis in Iran stems from decades of water-intensive development, fragmented governance and national priorities that sidelined environmental protection. Without major governance reform and reprioritization, technical solutions alone cannot stop worsening water and environmental degradation.

☐ DOI: <https://doi.org/10.1038/s41893-026-01784-0>

62. [ASAP] Role of Microoxic Control in Alternating Anoxic/Oxic Systems for Selective Enrichment of N₂O-Reducing Bacteria and N₂O Mitigation in Wastewater Treatment

【交替缺氧/好氧系统中微氧控制对选择性富集 N₂O 还原菌和废水处理 N₂O 减排的作用】

☐ Environmental Science & Technology | Thu, 19 Mar 2026 07: 【Latest】 | 重要性评分: 36

☐ Abstract:

Environmental Science & Technology DOI: 10.1021/acs.est.5c07873

☐ 中文摘要:

本研究揭示在交替缺氧/好氧废水处理系统中，精确的微氧环境控制可以选择性富集 N₂O 还原细菌，从而显著降低废水处理过程中强效温室气体 N₂O 的排放。

☐ 深度解读:

本研究发表于 Environmental Science & Technology，提出了一种通过微环境工程调控微生物群落来减少废水处理温室气体排放的创新策略。N₂O 的全球增温潜势是 CO₂ 的 265 倍，废水处理是重要的人

为 N₂O 排放源。通过精确调控溶解氧浓度至‘微氧’区间，可以定向培养 N₂O 还原菌，将这一强效温室气体彻底还原为 N₂。对中国城市污水处理厂的碳排放核算和减排技术升级具有实际应用价值。

□ DOI: <https://doi.org/10.1021/acs.est.5c07873>

63. Chronic dietary exposure and arsenic availability in tea agro-ecosystems: Insights from free ion activity model and sobol sensitivity analysis

□ Science of the Total Environment 【Latest】 | 重要性评分: 36

□ Abstract:

Publication date: 25 April 2026 Source: Science of The Total Environment, Volume 1027 Author(s): Saibal Ghosh, Piw Das, Bishwapran Kashyap, Sristi Das, Nazneen Hussain, Raktim Pal

□ DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726003669>

64. The critical role of soil moisture in compound hazards

【土壤湿度在复合灾害中的关键作用】

□ Nature Geoscience | 2026-03-13 【Latest】 | 重要性评分: 34

□ Abstract:

Nature Geoscience, Published online: 13 March 2026; doi:10.1038/s41561-026-01936-z Soil moisture drives critical land-atmosphere interactions and compound hazards. Closing gaps in data and models is key to integrating soil moisture into predictive frameworks for improved hazard forecasting.

□ 中文摘要:

本研究揭示土壤湿度在热浪-干旱、洪水-滑坡等复合灾害中发挥关键的‘连接枢纽’作用。土壤湿度异常既是前序灾害的结果，又是后续灾害的预置条件，使得多种极端事件相互耦合并放大。

□ 深度解读:

本研究发表于 Nature Geoscience，系统阐述了土壤湿度作为复合灾害‘耦合节点’的关键角色。在干旱-热浪复合事件中，土壤干燥通过减少蒸散冷却效应加剧地表增温；在暴雨-洪涝事件中，土壤含水量影响入渗与地表径流分配。这一机制性认识对极端天气预报和灾害风险评估具有重要改进潜力，对中国长江流域等复合灾害频发区域的防灾减灾具有直接应用价值。

□ DOI: <https://doi.org/10.1038/s41561-026-01936-z>

65. Effect of climate-driven childhood sleep erosion on potential regional economic inequality

【气候驱动的儿童睡眠侵蚀对区域经济发展潜力的影响】

□ Nature Sustainability | 2026-03-13 【Latest】 | 重要性评分: 34

□ Abstract:

Nature Sustainability, Published online: 13 March 2026; doi:10.1038/s41893-026-01779-x Hot climates erode children's sleep, threatening cognitive development with likely socioeconomic consequences. A global analysis finds potential links between future sleep losses and trillions of dollars in economic costs that disproportionately burden poorer regions.

□ 中文摘要:

本研究评估气候变暖导致的儿童睡眠损失对区域经济发展的潜在影响，通过量化睡眠不足-认知发展-人力资本-经济产出的级联路径，揭示气候变化对人类发展的深远经济代价。

□ 深度解读:

本研究发表于 Nature Sustainability，将气候-睡眠-经济发展创新性地连接，揭示了一条被忽视的气候变化影响人类发展的路径。儿童期睡眠不足影响大脑发育和学业表现，长期积累将损害区域人力资本质量和经济增长潜力。热带和亚热带发展中国家的儿童面临最大风险，这进一步加剧了全球不平等。对中国南方城市的儿童健康政策和气候适应投资决策具有启示意义。

□ DOI: <https://doi.org/10.1038/s41893-026-01779-x>

66. [ASAP] Specific Ion Chemistry at the Air – Water Interface of Nitrite/Nitrate-Containing Droplets

□ Environmental Science & Technology | Fri, 20 Mar 2026 05: **【Latest】** | 重要性评分: 34

□ Abstract:

Environmental Science & Technology DOI: 10.1021/acs.est.5c15074

□ DOI: <https://doi.org/10.1021/acs.est.5c15074>

67. On the importance of soil texture for predicting future global soil respiration

【土壤质地对预测未来全球土壤呼吸的重要性】

□ Science of the Total Environment **【Latest】** | 重要性评分: 34

□ Abstract:

Publication date: 25 April 2026 Source: Science of The Total Environment, Volume 1027 Author(s): Lucas Carvalho Gomes, Cássio Marques Moquedace, Ivan F. Souza, Ben Bond-Lamberty, Rodrigo Vargas, Lars Vesterdal, Gustavo V. Veloso, Marcio Rocha Francelino, Carlos E.G.R. Schaefer, Kendalynn A. Morris, Elpidio I. Fernandes-Filho

□ 中文摘要:

本研究揭示土壤质地（砂粒、粉粒、粘粒比例）是预测全球土壤呼吸（碳释放）的关键但被忽视的因子。纳入土壤质地信息可显著改善土壤碳排放的空间预测精度。

□ 深度解读:

本研究发表于 Science of the Total Environment，指出当前全球土壤碳模型中普遍忽视的一个关键参数——土壤质地。粘土含量高的土壤通过有机-矿物复合体保护碳免受微生物分解，而砂质土壤则碳损失更快。纳入土壤质地后模型预测精度显著提升，对改进 CMIP 系列地球系统模型中的土壤碳模块具有实际指导意义。

□ DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726003244>

68. Resilient adults but vulnerable larvae: demographic pathways of chiton decline under ocean acidification

【成体有韧性但幼体脆弱：海洋变暖下石鳖衰退的种群统计学路径】

□ Science of the Total Environment **【Latest】** | 重要性评分: 34

□ Abstract:

Publication date: 25 April 2026 Source: Science of The Total Environment, Volume 1027 Author(s): Ryo Matsumoto, Katsunori Kimoto, Koetsu Kon

□ 中文摘要:

本研究揭示在海洋变暖条件下，石鳖类的成体表现出较强的耐热性，但幼体阶段对升温高度敏感。这种生活史阶段差异化的脆弱性导致种群补充下降，是该类群长期衰退的关键种群统计学机制。

□ 深度解读:

本研究发表于 Science of the Total Environment，揭示了海洋无脊椎动物应对气候变化时一种常被忽视的脆弱性模式：‘成体耐受-幼体敏感’。这种生活史瓶颈效应意味着基于成体耐热性的风险评估会严重低估物种的真实气候脆弱性。对中国近海潮间带底栖生物保护和海洋保护区管理策略（需要关注繁殖期和幼体栖息地）具有重要启示。

□ DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726003323>

69. Modelling reveals redox budget transfer in Mariana-type subduction systems

□ Nature Geoscience | 2026-03-19 【Latest】 | 重要性评分: 32

□ Abstract:

Nature Geoscience, Published online: 19 March 2026; doi:10.1038/s41561-026-01937-y Subduction transfers redox budgets from the surface into the mantle. Quantitative modelling of a Mariana-type subduction system shows that mantle oxidation is dominated by sulfur-rich fluids and Fe³⁺-rich melts at sub-arc and back-arc depths, respectively. Most of the redox budget is ultimately transported into the deep mantle with the subducting slab.

□ DOI: <https://doi.org/10.1038/s41561-026-01937-y>

70. [ASAP] MXene-Based Nanoconfined Ion Channels for Ultratrace Detection of Short-Chain Per- and Polyfluoroalkyl Substance Trifluoromethanesulfonic Acid and Its On-site Application

□ Environmental Science & Technology | Fri, 20 Mar 2026 01: 【Latest】 | 重要性评分: 32

□ Abstract:

Environmental Science & Technology DOI: 10.1021/acs.est.5c15498

□ DOI: <https://doi.org/10.1021/acs.est.5c15498>

71. [ASAP] Hydrodynamic Forces as an Overlooked Driver of Phosphorus Mobilization via Piezoelectric Activation

□ Environmental Science & Technology | Fri, 20 Mar 2026 12: 【Latest】 | 重要性评分: 32

□ Abstract:

Environmental Science & Technology DOI: 10.1021/acs.est.5c17156

□ DOI: <https://doi.org/10.1021/acs.est.5c17156>

72. The evaluation to combined toxic effects of polypropylene microplastics and benzo[a]pyrene on macrophage cells

□ Science of the Total Environment 【Latest】 | 重要性评分: 32

□ Abstract:

Publication date: 25 April 2026 Source: Science of The Total Environment, Volume 1027 Author(s): Ye-xuan Zhang, Rui Hou, Si-si Chen, Xue-wen Guo, Hong-juan Chen, Li Mao, Wei-juan Zheng, Hong-zhen Lian

□ DOI: <https://www.sciencedirect.com/science/article/pii/S0048969726003700>

73. High-resolution models better simulate historical snowpack declines in the Upper Colorado River Basin

☐ Environmental Research Letters | 2026-03-17T00:00:00Z 【Latest】 | 重要性评分: 25

☐ Abstract:

The Colorado River supplies water to 40 million people, 4.5 million acres of irrigated land, seven US states, over two dozen federally recognized and sovereign tribes and Mexico. River discharge is strongly controlled by snowpack accumulation and melt in the headwater regions, making upper-basin snow dynamics a primary driver of water availability throughout the basin. Therefore, changes in snow water equivalent (SWE) are critical for water resources planning in the Upper Colorado River Basin (UCRB), yet there is considerable uncertainty even in historical SWE trends. We compare historical UCRB SWE trends across various gridded snow products, including new Weather Research and Forecasting simulations at 45, 9, and 3 km resolution, against snow telemetry station data. We find that SWE trend

☐ DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae4113>

74. Land use and land cover change intensified soil moisture drought: evidence from CMIP6-LUMIP

☐ Environmental Research Letters | 2026-03-17T00:00:00Z 【Latest】 | 重要性评分: 25

☐ Abstract:

Soil moisture drought (SMD) critically constrains terrestrial ecosystems and agricultural productivity, yet the long-term contribution of historical land use and land cover change (LULCC) to global drought trends remains insufficiently quantified. We quantify the impacts of historical LULCC on SMD characteristics (event number, duration, severity, intensity) over 1901–2014 using seven offline land surface simulations from the CMIP6 Land Use Model Intercomparison Project (LUMIP). LULCC intensifies SMD over more than half of the global land area, with the strongest increases in mid-latitude regions, particularly in central North America. The drought-affected area expanded throughout the twentieth century, with drought events becoming increasingly prolonged and severe. Diagnostic analyses of

☐ DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae4ca5>

75. Tracking shifts in European drought hotspots

☐ Environmental Research Letters | 2026-03-17T00:00:00Z 【Latest】 | 重要性评分: 23

☐ Abstract:

Drought is a complex hazard with far-reaching socio-economic and ecological impacts that are often inadequately captured by conventional indices based solely on hydroclimatic anomalies. To bridge this gap, this study develops novel impact-based combined drought indices (iCDIs) that directly relate hydroclimatic drivers to remotely sensed vegetation stress, here used as a proxy for observed agricultural impacts. The proposed framework employs a machine learning framework to optimize the selection of hydroclimatic predictors including precipitation, temperature, and soil moisture across distinct hydrological clusters of the European domain. This bottom-up machine learning-based approach outperforms traditional indices in reproducing observed vegetation responses across European river basins.

☐ DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae4ca7>

76. The effects of self-governance arrangements on the climate change resilience of small-scale fisheries in Mexico

☐ Global Environmental Change 【Latest】 | 重要性评分: 21

☐ Abstract:

Publication date: July 2026Source: Global Environmental Change, Volume 98Author(s): Xochitl Édua Elías Ilosvay, Sarah Elkin, Sebastian C.A. Ferse, Eréndira Aceves-Bueno, Javier Tovar-Ávila, Jhosafat Rentería-Bravo, Irving Alexis Medina Santiago, Jorge García Molinos, Elena Ojea

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S095937802600021X>

77. Do nature-based solutions deliver climate adaptation equally? Comparing mangroves and seawalls in the Global South

☐ Climatic Change | 2026-03-19 **【Latest】** | 重要性评分: 21

☐ DOI: <https://link.springer.com/article/10.1007/s10584-026-04151-2>

78. Extreme precipitation, exacerbated by anthropogenic climate change, drove Peru's record-breaking 2023 dengue outbreak

☐ One Earth **【Latest】** | 重要性评分: 19

☐ Abstract:

Publication date: Available online 17 March 2026Source: One EarthAuthor(s): Mallory J. Harris, Jared T. Trok, Kevin S. Martel, Mercy J. Borbor-Cordova, Noah S. Diffenbaugh, César V. Munayco, Andrés G. Lescano, Erin A. Mordecai

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S2590332226000205>

79. Climate change litigation and intersectionality in climate justice

☐ Global Environmental Change **【Latest】** | 重要性评分: 19

☐ Abstract:

Publication date: July 2026Source: Global Environmental Change, Volume 98Author(s): Elisabeth Schill, Emilie Greve Pobiega, Joyce Soo, Emily Boyd

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0959378026000208>

80. High-resolution data to detect meteorological sensitivity of health facilities' electricity systems in the Democratic Republic of the Congo

☐ Environmental Research Letters | 2026-03-20T00:00:00Z **【Latest】** | 重要性评分: 19

☐ Abstract:

Reliable electricity is essential for healthcare delivery, particularly in resource-constrained settings vulnerable to weather extremes. This study investigated the associations between temperature extremes and precipitation anomalies and power quality and reliability (PQR) in 24 healthcare facilities across North Kivu, Democratic Republic of the Congo. Using high-frequency sensor data and satellite-derived climate indices from May 2022 to October 2024, we assessed relationships between meteorological conditions (dry and wet extremes, heatwaves) and key PQR metrics: daily power source uptime, within-nominal voltage time, and within-nominal frequency time. Regression models incorporating monthly fixed effects revealed that small-scale hydroelectric systems, especially those with poor maintenance

☐ DOI: <https://iopscience.iop.org/article/10.1088/1748-9326/ae4b57>

81. How well do we report on expert elicitation for climate change decision support systems?

☐ Climatic Change | 2026-03-18 **【Latest】** | 重要性评分: 17

☐ DOI: <https://link.springer.com/article/10.1007/s10584-026-04143-2>

82. Unravelling the impact of changing rainy seasons on rice production in Southeast Asia: a climate change high-resolution perspective

☐ Climatic Change | 2026-03-16 **【Latest】** | 重要性评分: 17

☐ DOI: <https://link.springer.com/article/10.1007/s10584-026-04136-1>

83. Seasonal agricultural vulnerability in semi-arid Morocco: combining remote sensing and farmer knowledge to inform climate adaptation

☐ Climatic Change | 2026-03-19 **【Latest】** | 重要性评分: 13

☐ DOI: <https://link.springer.com/article/10.1007/s10584-026-04160-1>

84. Decadal prediction system based on the INM RAS climate model

☐ Climate Dynamics | 2026-03-16 **【Latest】** | 重要性评分: 12

☐ Abstract:

Based on the INMCM5 coupled general circulation model, the decadal climate prediction system has been developed. The system provides a complete cycle of decadal predictions from preparation of initial conditions to prognostic ensemble calculations and analysis of the results. The system has been evaluated for the years 1960–2020. The quality of INMCM5 hindcasts for annual and 5-year mean global and regional near-surface air temperature, precipitation, and sea level pressure as well as timeseries of key climate indices is on par with the mean level of the World Meteorological Organization Lead Centre for Annual-to-Decadal Climate Prediction (WMO LC-ADCP) models. Besides estimates recommended by the WMO LC-ADCP, an assessment of the East Atlantic-West Russian Oscillation, Western Pacific Osc

☐ DOI: <https://link.springer.com/article/10.1007/s00382-026-08047-w>

85. [ASAP] Climate-Driven Challenges for Production of Clean and Safe Potable Water: Concerns for Dissolved Organic Matter and Anthropogenic Pollution

☐ ES&T Letters | Wed, 18 Mar 2026 04: **【Latest】** | 重要性评分: 10

☐ Abstract:

Environmental Science & Technology Letters DOI: 10.1021/acs.estlett.5c00310

☐ DOI: <https://doi.org/10.1021/acs.estlett.5c00310>

86. A global perspective: quantifying disturbance and reclamation of surface coal mining through remote sensing innovations

☐ Global Environmental Change **【Latest】** | 重要性评分: 9

☐ Abstract:

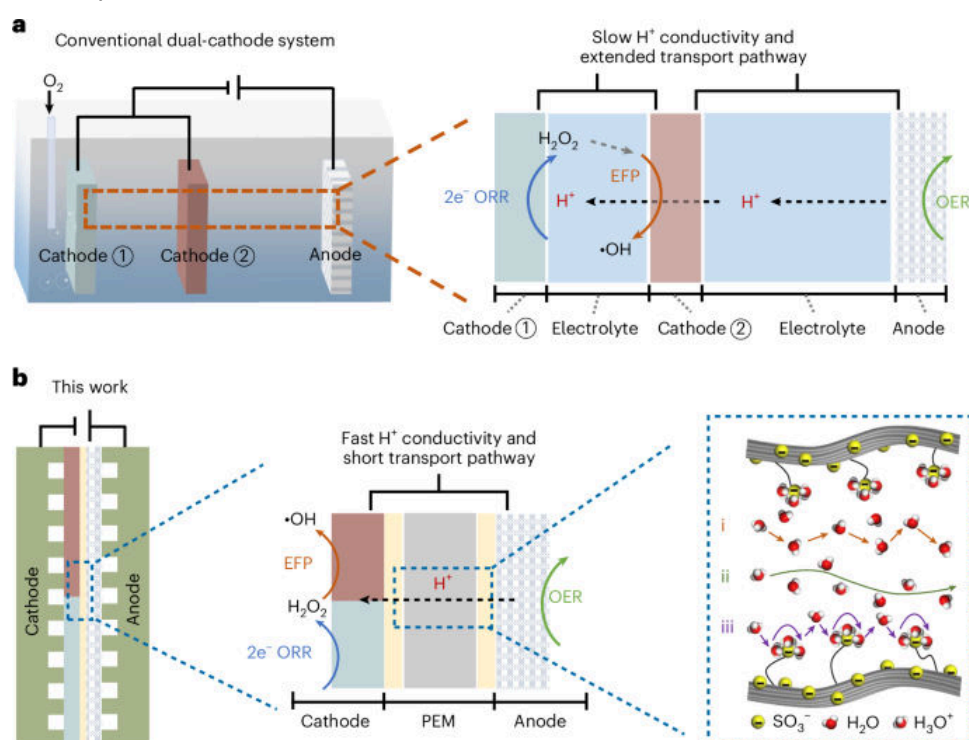
【海洋与水生】

87. Molecular oxygen cascade reduction to $\cdot\text{OH}$ via coplanar dual-electrocatalytic zone achieving electrolyte-free water purification

【共面双电催化区实现分子氧级联还原生成 $\cdot\text{OH}$ 的无电解质水净化】

□ Nature Water | 2026-03-10 【Latest】 | 重要性评分: 36

□ Graphical Abstract:



□ Abstract:

Nature Water, Published online: 10 March 2026; doi:10.1038/s44221-026-00606-zThis study presents an electrolyte-free electrochemical system that leverages dual electrocatalytic zones to effectively remove a wide range of recalcitrant pollutants across various water matrices, including low-conductivity organic wastewater.

□ 中文摘要:

本研究提出了一种无需添加电解质的电化学水处理新体系，通过双电催化区的协同作用将分子氧级联还原为羟基自由基 ($\cdot\text{OH}$)，实现对多种难降解污染物的高效去除。

□ 深度解读:

本研究发表于 Nature Water，突破了传统电化学水处理必须添加电解质的技术瓶颈。无电解质运行意味着不产生二次污染，大幅简化了水处理工艺。该技术对从药品残留到全氟化合物等多种新污染物均有效，代表了水处理电化学技术的重要范式转变。对中国饮用水深度处理和工业废水新污染物治理具有技术应用前景。

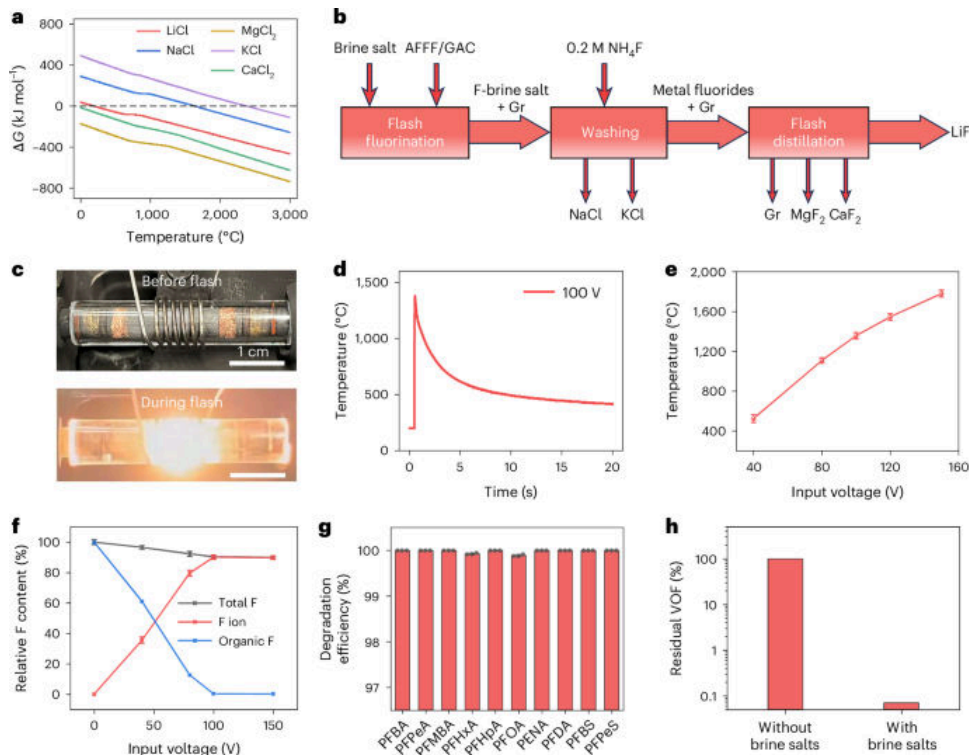
□ DOI: <https://doi.org/10.1038/s44221-026-00606-z>

88. Waste per- and polyfluoroalkyl substance-assisted flash fluorination for lithium recovery from brine

【利用废弃全氟/多氟烷基物质辅助闪蒸氟化从卤水中回收锂】

□ Nature Water | 2026-03-10 【Latest】 | 重要性评分: 36

□ Graphical Abstract:



□ Abstract:

Nature Water, Published online: 10 March 2026; doi:10.1038/s44221-026-00593-1 An electrothermal fluorination methodology has been developed to transform granular activated carbon-sorbed aqueous film-forming foam waste into graphene and a fluorinating reagent, subsequently enabling lithium recovery from brine sources.

□ 中文摘要:

本研究开发了一种电热氟化方法，将活性炭吸附的含氟表面活性剂废物转化为石墨烯和氟化剂，用于从卤水中高效选择性回收锂。同时实现了 PFAS 废物处理和锂资源回收的双重目标。

□ 深度解读:

本研究发表于 Nature Water，巧妙地将两个环境难题——PFAS（‘永久化学品’）污染治理和锂资源短缺——转化为一个协同解决方案。通过超高温闪蒸处理，将 PFAS 废物转化为石墨烯和氟化试剂，后者可用于从盐湖卤水中高效提锂。这一‘以废治废’的循环经济路径对中国青海、西藏盐湖锂资源开发和 PFAS 污染治理具有重要技术启发。

□ DOI: <https://doi.org/10.1038/s44221-026-00593-1>

89. Periphyton communities in the drifting water hyacinth habitats of Cochin backwaters- South-west coast of India

□ Estuarine Coastal and Shelf Science 【Latest】 | 重要性评分: 34

□ Abstract:

Publication date: 15 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 335Author(s): K.H. Afritha, K.S. Amal, Renju Mohan, Aishwarya Purushothaman, Lathika Cicily Thomas, P. Ajayakumar, K.B. Padmakumar

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426001150>

90. Organic carbon distribution in surface sediments of northern South China Sea: coastal carbon pump with no large-river influence

☐ Estuarine Coastal and Shelf Science **【Latest】** | 重要性评分: 34

☐ Abstract:

Publication date: 15 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 335Author(s): Wenpeng Li, Jinpeng Wang, Huawei Wang, Hongwei Chen, Xiaohui Zhai, Zhongbo Wang, Xinxin Li

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426001332>

91. Microplastic distribution and occurrence in benthic species along river–sea continua: Implications for coastal aquaculture in the South Yellow Sea

☐ Estuarine Coastal and Shelf Science **【Latest】** | 重要性评分: 32

☐ Abstract:

Publication date: 15 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 335Author(s): Xiaoping Sun, Ruiping Yang, Jing Ji, Jason C. White, Yu Shen

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426001228>

92. Planktonic foraminiferal assemblages across a mid–outer shelf of the southern Red Sea, Off Yemen

☐ Estuarine Coastal and Shelf Science **【Latest】** | 重要性评分: 32

☐ Abstract:

Publication date: 15 July 2026Source: Estuarine, Coastal and Shelf Science, Volume 335Author(s): Youssef S. Bazeen, Reem M. Muaamar, Mohammed A. Al-Wosabi, Abdel Galil A. Hewaidy, Abdulkarim A. Al-Subbary

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426001216>

93. Interfacial interaction-driven rapid capture and on-site analysis of nano- and microplastics enabled by multifunctional magnetic adsorbent

☐ Nature Water | 2026-03-13 **【Latest】** | 重要性评分: 32

☐ Abstract:

Nature Water, Published online: 13 March 2026; doi:10.1038/s44221-026-00610-3A multifunctional adsorbent, composed of copper-doped polydopamine-functionalized magnetic silica, enables the simultaneous removal and on-site detection of label-free nano- and microplastics.

☐ DOI: <https://doi.org/10.1038/s44221-026-00610-3>

94. Steering the nitrate electroreduction pathway via nanoconfinement-induced hydrogen-bond network regulation

☐ Nature Water | 2026-03-11 **【Latest】** | 重要性评分: 32

☐ Abstract:

Nature Water, Published online: 11 March 2026; doi:10.1038/s44221-026-00600-5 Electrochemical nitrate reduction can simultaneously remove pollutants and produce ammonia but is often limited by hydrogen side reactions. This study shows that nanoconfined CuCo catalysts enable a proton-coupled electron transfer pathway that delivers highly selective, efficient and stable ammonia synthesis.

☐ DOI: <https://doi.org/10.1038/s44221-026-00600-5>

95. Water scientists must become agenda-setters

☐ Nature Water | 2026-03-09 **【Latest】** | 重要性评分: 32

☐ Abstract:

Nature Water, Published online: 09 March 2026; doi:10.1038/s44221-026-00613-0 The global water agenda is outdated and narrow and is framed mainly as a downstream impact sector. Scientists must step up to help the world recognize water as an opportunity sector and to design a bolder water agenda.

☐ DOI: <https://doi.org/10.1038/s44221-026-00613-0>

96. Spatio-temporal dynamics and socioeconomic drivers of seaport expansion in China from 1990 to 2024

☐ Estuarine Coastal and Shelf Science **【Latest】** | 重要性评分: 30

☐ Abstract:

Publication date: 15 July 2026 Source: Estuarine, Coastal and Shelf Science, Volume 335 Author(s): Tianhao Lian, Yuanheng Sun, Senlin Teng, Xueyuan Zhu, Ying Li

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0272771426001319>

97. Carry-over effects of decreased salinity on larval metamorphosis and the early juvenile in context with eReefs hydrodynamic models

☐ Coral Reefs | 2026-03-18 **【Latest】** | 重要性评分: 27

☐ Abstract:

The environmental conditions experienced by marine invertebrate larvae can have carry-over effects on later life stage performance, affecting population dynamics. For crown-of-thorns sea stars (CoTS; *Acanthaster* spp.), the terrestrial runoff hypothesis (TRH) suggests that eutrophic runoff plumes boost larval phytoplankton food supply, increasing metamorphic success and seeding outbreaks. However, river plumes also decrease salinity which can be deleterious to sea star larvae. We investigated the impact of decreased salinity (17–30 PSU) across a range of exposure durations on survival, normal development, and the ability of settlement-ready brachiolaria larvae to produce a juvenile. Salinity performance curves identified salinity levels with deleterious effects and the tipping points for 50

☐ DOI: <https://link.springer.com/article/10.1007/s00338-026-02819-w>

98. Acidification and deoxygenation matter in assessing redistribution of global cold-water coral biodiversity induced by climate change

☐ Limnology and Oceanography | Wed, 18 Mar 2026 06: **【Latest】** | 重要性评分: 25

☐ DOI: <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.70351>

99. Herbivores in hot water: thermal limits and avoidance of marine heatwave conditions by *Acanthurus triostegus*

☐ Coral Reefs | 2026-03-20 **【Latest】** | 重要性评分: 24

☐ Abstract:

Herbivorous fishes play a vital role in maintaining coral reef health and dominance by mitigating algal overgrowth. However, the functional traits of these fishes are increasingly threatened by marine heatwaves and rising ocean temperatures that cause coral bleaching, as these encroach on the thermal tolerance limits of sensitive species and drive some to vacate thermally stressed habitats. Using a controlled laboratory setup, we assessed the critical thermal maximum (CT_{max}), temperature preference (T_{pref}), and avoidance temperature (T_{avoid}) of a common herbivorous coral reef fish *Acanthurus triostegus* (manini) under seasonal (winter = 24.0 ± 0.1 °C, summer = 27.5 ± 0.1 °C) and marine heatwave conditions (31.0 ± 0.1 °C). CT_{max} increased significantly under marine heatwave conditions (from

☐ DOI: <https://link.springer.com/article/10.1007/s00338-026-02853-8>

100. Return of the lamingtons—boom-and-bust populations of the barrens forming sea urchin *Tripneustes australiae* in Lord Howe Island Marine Park

☐ Coral Reefs | 2026-03-20 **【Latest】** | 重要性评分: 20

☐ Abstract:

Sea urchins in the genus *Tripneustes* have fast growth and boom-and-bust population dynamics. *Tripneustes australiae* is endemic to southeast Australia and northwest New Zealand and in 2024 exhibited a population increase ('boom' phase) in Lord Howe Island Marine Park (LHIMP). Grazing pressure in population outbreaks of this species creates barrens-type habitat. We used Reef Life Survey data to document the density of *Tripneustes* species spanning 18 years (2006–2024). Two outbreaks were evident, the first in 2008 followed by a marked decline and the second in 2024. There was no difference in the densities of *T. australiae* in no-take sanctuary zones and habitat protection zones of LHIMP over most years, except for higher densities in sanctuary zones in the boom year 2024. The Admiralty Island

☐ DOI: <https://link.springer.com/article/10.1007/s00338-026-02851-w>

101. Event-driven monitoring reveals multi-pesticide contamination of small standing waters beyond adjacent de facto farming practice risks

☐ Hydrobiologia | 2026-03-20 **【Latest】** | 重要性评分: 19

☐ Abstract:

Agricultural practices are a major source of pesticide contamination in freshwater ecosystems. Lentic small water bodies (LSWBs), which make up the vast majority of freshwater systems, are particularly vulnerable to various pesticide entry routes, including surface runoff, spray drift and subsurface flow from adjacent agricultural land. This study aimed to assess whether event-driven monitoring could effectively detect pesticide contamination in LSBWs. Using 1 year of event-driven monitoring data from three LSBWs, we compared measured pesticide concentrations with predicted values from the SYNOPSIS-WEB risk assessment tool. The results revealed notable discrepancies.

While SYNOPSIS-WEB predicted certain contamination events, on-site monitoring frequently showed the presence of pesticides that

☐ DOI: <https://link.springer.com/article/10.1007/s10750-026-06130-1>

102. Deep Learning Methods for detecting marine vertebrates and invertebrates from satellite, aerial, and underwater imagery: A review

☐ Deep Sea Research I **【Latest】** | 重要性评分: 18

☐ Abstract:

Publication date: June 2026 Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229 Author(s): Arthur Benard, Hervé Nikue Amassah, Guillaume Feuilloley

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0967063726000312>

103. Agricultural landscapes decrease zooplankton hatching rates in temporary ponds

☐ Hydrobiologia | 2026-03-20 **【Latest】** | 重要性评分: 18

☐ Abstract:

We investigated the role of dormant zooplankton in maintaining biodiversity in temporary environments of semi-arid regions under different levels of anthropogenic pressure: high (intensive agriculture with pesticide use), moderate (livestock), and low (preserved areas). Samples were collected and the experiment was conducted in the field over 31 days, with collections performed every two days, revealing distinct hatching patterns among environments. Hatching rates and diversity were lower in highly impacted areas. Differences between active and dormant communities were notable, with reduced richness and diversity under strong anthropogenic influence. Conchostracans, anostracans, and testate amoebae emerged within the first 48 h, especially in preserved areas. Microcrustaceans such as clado

☐ DOI: <https://link.springer.com/article/10.1007/s10750-026-06165-4>

104. Inferring the distribution of octocorals *Nanipora* and *Heliopora* along the coast of the Kerama Islands, Okinawa, Japan, by eDNA metabarcoding

☐ Coral Reefs | 2026-03-20 **【Latest】** | 重要性评分: 17

☐ Abstract:

Two zooxanthellate octocorals, *Heliopora coerulea* and *Nanipora kamurai* (family Helioporidae, order Scleractinia, class Octocorallia), produce aragonite calcium carbonate skeletons similar to those of scleractinian hexacorals and contribute to the formation of complex coral reef ecosystems. *H. coerulea* is known to occur along the coasts of the Kerama Islands, Okinawa, Japan, and *N. kamurai* was originally described from the Kerama Islands. An environmental DNA metabarcoding (eDNA-M) method has recently been introduced for comprehensive surveys of zooxanthellate scleractinian corals, supplementing conventional visual census by SCUBA diving or snorkeling. In particular, our recently established Scl-eDNA-M (Scleractinia environmental DNA Metabarcoding) system, which targets mitochondrial 12S

☐ DOI: <https://link.springer.com/article/10.1007/s00338-026-02857-4>

105. Moderate nutrients and future warmer wintertime temperatures have a net positive effect on the physiology of three species of coral from the southern Great Barrier Reef

☐ Coral Reefs | 2026-03-20 **【Latest】** | 重要性评分: 16

☐ Abstract:

High nutrient concentrations as well as elevated temperature above summer mean monthly maximums have significant negative impacts on the physiology of the coral holobiont. However, moderate nutrient concentrations ($< 6 \mu\text{mol L}^{-1} \text{NO}_3^-$ and $0.40 \mu\text{mol L}^{-1} \text{PO}_4^{3-}$) have been shown to have a positive effect on coral and endosymbiont physiology and to minimize or mitigate the negative effects of summertime elevated temperature. As the oceans continue to warm, corals will face elevated temperatures in winter as well as summer. Yet, few studies have assessed the potential impacts of increases in winter temperature on corals and none have evaluated the possible mitigating or exacerbating effect of nutrients during warmer winters of the future. Here, we tested the effects of elevated temperature and moder

☐ DOI: <https://link.springer.com/article/10.1007/s00338-026-02854-7>

106. Taxonomy and Functional Traits Modulate the Environmental Niche Dimensions of Crustacean Zooplankton Communities in Submerged Aquatic Vegetation Beds

☐ Freshwater Biology | Sun, 08 Mar 2026 06: **【Latest】** | 重要性评分: 14

☐ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/fwb.70193>

107. Long-term fertilization regimes mitigate dissolved organic matter leaching and alter its chemodiversity in agricultural soil leachate

☐ Water Research **【Latest】** | 重要性评分: 13

☐ Abstract:

Publication date: 1 June 2026 Source: Water Research, Volume 297 Author(s): Longlong An, Shuqin He, Xin Song, Zicheng Zheng, Tingxuan Li, Le Han

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0043135426004446>

108. Some like it cold: the effects of diapause conditions on reactivation of ephippial eggs of *Daphnia magna*

☐ Hydrobiologia | 2026-03-20 **【Latest】** | 重要性评分: 12

☐ Abstract:

Survival of aquatic invertebrates relies on the resistance of dormant stages to environmental stress and the synchronization of dormancy with seasonal changes. We investigated hatching patterns in *Daphnia magna* from four Mediterranean populations that experience seasonal drying. Ephippial eggs were either transferred directly to hatching-stimulation conditions or first exposed to one of the following storage treatments: wet, dark storage at 20°C or 1°C ; or dry, dark storage at 20 , 40 , or 60°C . Surprisingly, the most successful and synchronous hatching occurred after 2-month dark, cold (1°C) storage at the temperature that rarely, if ever, occurs in their natural habitats. Dry storage led to more gradual hatching. Early diapause termination was observed in two of the studied populations

☐ DOI: <https://link.springer.com/article/10.1007/s10750-026-06152-9>

109. Reconstructing missing marine CDOM profiles using environmental covariates and graph-based deep learning

☐ Deep Sea Research I **【Latest】** | 重要性评分: 11

☐ Abstract:

Publication date: June 2026Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229Author(s): Beibei Xie, Xinhui Liu, Fanzhi Meng, Di Zhang, Kailong Yuan

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0967063726000300>

110. Multibeam seabed bathymetry of the Long-Term Ecological Research observatory HAUS-GARTEN (Fram Strait)

☐ Deep Sea Research II 【Latest】 | 重要性评分: 11

☐ Abstract:

Publication date: June 2026Source: Deep Sea Research Part II: Topical Studies in Oceanography, Volume 227Author(s): Boris Dorschel, Simon Dreutter, Lilian Boehringer

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0967064526000305>

111. Microplastics abundance in marine sediments and stomach contents of benthic organisms

☐ Hydrobiologia | 2026-03-20 【Latest】 | 重要性评分: 10

☐ Abstract:

Microplastics are an emerging contaminant in marine ecosystems worldwide. This study evaluated the abundance of microplastics, their relationship with sediment type, and their ingestion by macroinvertebrates in the Eastern South Pacific Ocean and the Southern Caribbean Sea during the rainy and dry seasons. A granulometric analysis was performed applying a Pearson correlation matrix, and microplastics were extracted using the density flotation method and analyzed by type, color, and size. There was an average of 165 ± 22 microplastics/kg in the Eastern South Pacific Ocean and 296 ± 167 microplastics/kg in the Southern Caribbean Sea, however, there were no significant differences among regions or seasons. A negative relationship was found between microplastic abundance and coarser sediment,

☐ DOI: <https://link.springer.com/article/10.1007/s10750-026-06155-6>

112. Fine-Scale Genomic Divergence in the Amazonian Pirarucu (*Arapaima gigas*) Highlights the Need for Local Management Strategies

☐ Freshwater Biology | Sun, 15 Mar 2026 20: 【Latest】 | 重要性评分: 9

☐ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/fwb.70187>

113. Identifying Purse Seine Vessel Operations Through Machine Learning Models for Better Spatial Fishing Effort Estimates

☐ Fish and Fisheries | Fri, 13 Mar 2026 21: 【Latest】 | 重要性评分: 9

☐ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/faf.70078>

【保护与生物多样性】

114. Strengthening biodiversity conservation and One Health through ranger monitoring of wildlife health in protected areas

【通过保护区巡护员监测野生动物健康来加强生物多样性保护和‘同一健康’】

☐ Conservation Biology | Thu, 19 Mar 2026 00: **【Latest】** | 重要性评分: 40

☐ 中文摘要:

本研究探讨了保护区巡护员在监测野生动物健康中的作用, 以及如何通过这一基层监测网络同时推进生物多样性保护和'同一健康'(One Health)目标。研究提出了一个整合巡护员监测数据的框架, 将野生动物健康监测与疾病预警相连接。

☐ 深度解读:

本研究发表于 Conservation Biology, 提出了一个创新性的框架: 利用全球数以万计的保护区巡护员作为野生动物健康的'前哨监测网络'。巡护员日常巡护中收集的动物健康数据(异常行为、疾病症状、死亡事件)经过标准化后可构建疫病早期预警系统, 将生物多样性保护与传染病防控(One Health)有机整合。这对中国自然保护地体系改革和野生动物疫源疫病监测网络完善具有直接实践价值。

☐ DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70241>

115. A practical, adaptive compliance management framework for improving marine protected area effectiveness

【提高海洋保护区有效性的实用适应性合规管理框架】

☐ Conservation Biology | Wed, 18 Mar 2026 08: **【Latest】** | 重要性评分: 36

☐ 中文摘要:

本研究提出了一个实用的适应性合规管理框架, 用于提高海洋保护区的管理有效性。框架整合了执法监测、违规风险评估和利益相关者参与, 为保护区管理者提供了系统性的合规改进路径。

☐ 深度解读:

本研究发表于 Conservation Biology, 提出了一个可操作的海洋保护区合规管理框架。全球约 30% 的海洋保护区面临'纸上公园'问题(划定但缺乏有效管理), 该框架通过建立违规行为的风险评估矩阵和适应性执法策略, 为提升保护区实际保护效果提供了工具箱。对中国南海、东海海洋保护区的规范化管理和即将推进的海洋保护区扩展目标具有直接实践指导。

☐ DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70246>

116. Contrasting vulnerability profiles for insular assemblages of birds and mammals facing global change

【岛屿鸟类和哺乳动物群落在全球变化下的差异脆弱性】

☐ Conservation Biology | Sat, 14 Mar 2026 04: **【Latest】** | 重要性评分: 36

☐ 中文摘要:

本研究比较了面对全球变化(气候变化、入侵物种、栖息地丧失)时, 岛屿鸟类和哺乳动物群落表现出截然不同的脆弱性格局, 揭示了类群特异性保护需求。

☐ 深度解读:

本研究发表于 Conservation Biology, 系统比较了全球岛屿生态系统中鸟类和哺乳动物对多重全球变化驱动因素的差异化响应。发现鸟类和哺乳动物在气候变化适应能力、入侵物种抵抗力等方面存在系统性差异, 对岛屿保护区管理策略制定具有重要指导意义。对中国南海岛屿的生物多样性保护管理具有直接参考价值。

☐ DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70261>

117. Using resurvey data to predict changes in ecosystem functioning across protected and unprotected coastal dunes

【利用复查数据预测保护区生态系统功能变化】

☐ Conservation Biology | Sat, 14 Mar 2026 04: 【Latest】 | 重要性评分: 34

☐ 中文摘要:

本研究利用保护区内的长期重复调查（resurvey）数据，建立了预测生态系统功能变化的模型框架。通过分析群落组成和功能性状的时间变化，可以推断保护区内碳固存、养分循环等关键生态系统功能的变化趋势。

☐ 深度解读:

本研究发表于 Conservation Biology，提出了一种利用已有生物多样性调查数据评估保护区生态系统功能状态的新方法。传统保护效果评估多关注物种数量变化，而该框架通过功能性状将群落变化翻译为生态系统服务变化，填补了‘保护区保护了多少生态功能’这一关键评估缺口。对中国国家公园体系的保护成效评估机制完善具有方法论参考价值。

☐ DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70256>

118. Stepwise evolutionary transitions in Cladophorales plastid genomes reveal origins of hair-pin chromosomes

☐ Current Biology 【Latest】 | 重要性评分: 34

☐ Abstract:

Publication date: Available online 18 March 2026
Source: Current Biology
Author(s): Saelin Bjornson, Trevor T. Bringloe, Kavitha Uthannumallian, Ricardo De Paoli-Iseppi, Michael B. Clark, Chiel M.C. Cremen, Riyadh Hossen, Owen McGinley, Tara Jalali, John West, Frederik Leliaert, Richard Wetherbee, Olivier De Clerck, Sebastian Duchene, Jacob L. Steenwyk, Heroen Verbruggen

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226002253>

119. Regulatory compliance and support for altered management of the exotic pet trade

☐ Conservation Biology | Thu, 19 Mar 2026 03: 【Latest】 | 重要性评分: 32

☐ DOI: <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70257>

120. Myosin II-driven contractions of supporting cap cells promote sensory adaptation of Drosophila proprioceptors

☐ Current Biology 【Latest】 | 重要性评分: 32

☐ Abstract:

Publication date: Available online 17 March 2026
Source: Current Biology
Author(s): Chonglin Guan, Kengo Nishi, Christian T. Kreis, Oliver Bäümchen, Martin C. Göpfert, Christoph F. Schmidt

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S096098222600223X>

121. Recent experience and internal state shape local search strategies in flies

☐ Current Biology 【Latest】 | 重要性评分: 30

☐ Abstract:

Publication date: Available online 18 March 2026Source: Current BiologyAuthor(s): Dennis Goldschmidt, Yipei Guo, Shivam S. Chitnis, Christina Christoforou, Daniel B. Turner-Evans, Carlos Ribeiro, Ann M. Hermundstad, Vivek Jayaraman, Hannah Haberkern

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226001648>

122. Arp2/3-dependent actin assembly shapes endosomes and promotes intracellular trafficking in fission yeast

☐ Current Biology 【Latest】 | 重要性评分: 30

☐ Abstract:

Publication date: Available online 17 March 2026Source: Current BiologyAuthor(s): Alejandro Melero, Miguel Basante-Bedoya, Olivia Muriel-Lopez, Sophie G. Martin

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226002204>

123. Simultaneous path-integration recalibration in head direction and place cells

☐ Current Biology 【Latest】 | 重要性评分: 30

☐ Abstract:

Publication date: Available online 17 March 2026Source: Current BiologyAuthor(s): Ravikrishnan P. Jayakumar, Yotaro Sueoka, Marissa K. Ferreyros, Brian Y. Li, Manu S. Madhav, Xingyu Chen, James J. Knierim, Noah J. Cowan

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0960982226002228>

124. Patterns of phylogenetic structure across plant life-forms in a large temperate forest and their possible drivers

☐ Biodiversity and Conservation | 2026-03-12 【Latest】 | 重要性评分: 27

☐ Abstract:

Understanding the phylogenetic structure patterns of forest plants and their drivers is crucial for preserving the evolutionary potential of biodiversity. However, current knowledge remains relatively limited regarding high-latitude temperate forests. Here, we investigated over 38,000 plots in the largest natural forest region of China. Phylogenetic structure metrics (standardized phylogenetic diversity, SesPD; standardized mean pairwise distance, SesMPD; standardized mean nearest taxon distance, SesMNTD) were calculated. Geoclimatic factors (mean annual temperature, MAT; mean annual precipitation, MAP; altitude), community attributes (tree height, TH; diameter at breast height; shrub coverage; shrub abundance; herb coverage, HC), and soil property (total phosphorus) were derived from inte

☐ DOI: <https://link.springer.com/article/10.1007/s10531-026-03308-w>

125. Climate change – mediated catastrophe exacerbates genomic vulnerability of a coastal cycad

☐ Biological Conservation 【Latest】 | 重要性评分: 24

☐ Abstract:

Publication date: May 2026 Source: Biological Conservation, Volume 317 Author(s): Jui-Tse Chang, Koh Nakamura, Pei-Wei Sun, José Said Gutiérrez-Ortega, Min-Xin Luo, Chien-Ti Chao, Shao-Wei Ho, Yi Sun, Pei-Chun Liao

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0006320726000856>

126. Changes in South American migratory sheldgeese distribution associated with climate change

☐ Biological Conservation 【Latest】 | 重要性评分: 19

☐ Abstract:

Publication date: May 2026 Source: Biological Conservation, Volume 317 Author(s): Natalia A. Cossa, Nicolás A. Lois, Sergio A. Lambertucci

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0006320726000972>

127. Assessing vertical stratification of bats and nocturnal insects to infer potential collision risk at wind turbines in central Thailand

☐ Biodiversity and Conservation | 2026-03-14 【Latest】 | 重要性评分: 16

☐ Abstract:

The global growth of wind energy can harm wildlife and ecosystems, especially airborne animals, which face high collision risks with turbine blades. While detailed knowledge of bird and bat collision risk exists for temperate regions, risk assessments for insects and bats in the Global South remain lacking. To fill this gap, we investigated the activity of bats and nocturnal insects in Thailand at different heights above the ground. To this end, we suspended ultrasonic detectors and sticky traps from a helium-filled balloon kite at 25, 50, 75, and 100 m above ground, and also mounted them on a 2 m post to record ground-level activity. We documented the rotor-swept zone of local wind turbines to range between 21 and 159 m above ground. Mostly bat species from the guild of open-space forager

☐ DOI: <https://link.springer.com/article/10.1007/s10531-026-03307-x>

128. The role of an endangered vine parasite in promoting agritourism mediated conservation in Sabah, Malaysia

☐ Biodiversity and Conservation | 2026-03-13 【Latest】 | 重要性评分: 16

☐ Abstract:

Agritourism is increasingly viewed as an anchor for sustainable development in Sabah, Malaysia, extending beyond its well-known and established wildlife-based tourism industry. This commentary considers a future role of agritourism that, when combined with targeted conservation practices, has the potential to aid in the broader effort to protect Sabah's rich biodiversity and support local indigenous economies. While Rafflesia, which is frequently located on private land, is an iconic species that has long attracted international visitors, many regions in Sabah possess untapped potential for eco-focused agritourism within natural landscape settings. Agritourism that protects forest could also act as an impetus to engage in the re-wilding of areas via the re-introduction of declining and rare

☐ DOI: <https://link.springer.com/article/10.1007/s10531-026-03315-x>

129. Moth, beetle, and bug assemblages are shaped differently across a tropical rainforest vertical gradient

□ Biodiversity and Conservation | 2026-03-17 【Latest】 | 重要性评分: 12

□ Abstract:

Tropical forests harbour a large proportion of Earth's insect biodiversity. Tropical forest structure produces layers of habitat from the cooler, wetter understorey to the hotter, drier canopy, which provide stratified resources for inhabitants. This gradient of biotic and abiotic factors results in vertically stratified insect communities, often with different assemblages in the understorey and canopy. Vertical distributions of insect taxa are shaped by their niche requirements, with narrower vertical ranges for species with more specialised life history strategies and abiotic requirements. Many studies of vertical stratification focus on single taxonomic orders, often overlooking less well-worked orders, such as Hemiptera, and comparisons of the responses of different taxa sampled at the

□ DOI: <https://link.springer.com/article/10.1007/s10531-026-03274-3>

130. Combining Population Genomics With Ancient DNA to Understand Island Colonisation History of the Madagascar Turtle Dove

□ Molecular Ecology | Mon, 16 Mar 2026 23: 【Latest】 | 重要性评分: 9

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/mec.70299>

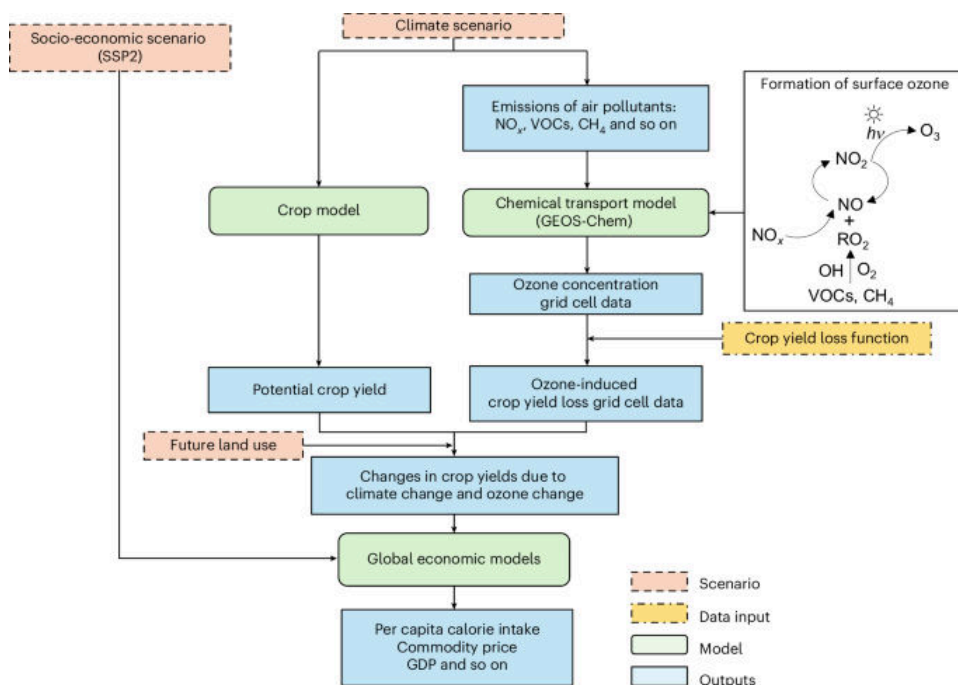
【植物与农业】

131. Ozone pollution reduction partially offsets the negative impact of climate change mitigation efforts on global hunger

【臭氧污染减少部分抵消气候减缓措施对全球饥饿的负面影响】

□ Nature Food | 2026-03-16 【Latest】 | 重要性评分: 51

□ Graphical Abstract:



☐ Abstract:

Nature Food, Published online: 16 March 2026; doi:10.1038/s43016-026-01322-3A multi-model intercomparison using six global agro-economic models explores how ozone reductions arising from climate change mitigation could impact crop yields. Yield changes are projected to 2050, linking reduced ozone to food security and the alleviation of hunger.

☐ 中文摘要:

通过六个全球农业经济模型的多模型比较研究，探讨气候变化减缓行动带来的臭氧浓度降低如何影响农作物产量。研究预测到 2050 年的产量变化，将臭氧减少与农业生产力和粮食安全相关联，揭示气候政策的粮食安全协同效益。

☐ 深度解读:

本研究发表于 Nature Food，采用多模型集成方法，首次系统量化了气候减缓政策通过降低地表臭氧浓度对全球农业产量和粮食安全的间接协同效益。研究发现减少化石燃料燃烧不仅可直接降低温室气体排放，还通过减少臭氧前体物排放改善作物生长环境，预计可使主要粮食作物产量提升 1-5%。这一双重效益对中国等农业大国尤为重要，为推进‘双碳’政策提供了额外的粮食安全协同论据。

☐ DOI: <https://doi.org/10.1038/s43016-026-01322-3>

132. Restoring the mana of the moana for ecological and food-system resilience in the Hauraki Gulf

【恢复豪拉基湾海洋的‘神力’以实现生态和食物系统韧性】

☐ Nature Food | 2026-03-19 【Latest】 | 重要性评分: 38

☐ Abstract:

Nature Food, Published online: 19 March 2026; doi:10.1038/s43016-026-01324-1Food-system resilience requires more than ecological protection—it must also centre Indigenous peoples’rights, customs and leadership. With genuine Māori co-governance, the Hauraki Gulf/Tikapa Moana Marine Protection Act 2025 could become a global model for integrating Indigenous peoples’food-system justice into marine conservation.

☐ 中文摘要:

食物系统韧性不仅需要生态保护，还必须以原住民权利、习俗和领导力为中心。在毛利族共同治理下，新西兰豪拉基湾的海洋修复可以同时实现生态恢复和食物主权。

☐ 深度解读:

本研究发表于 Nature Food，提出了将原住民传统生态知识与现代保护科学相融合的海洋修复范式。毛利族对海洋的‘mana’（神力/尊严）概念超越了西方生态学的功能性框架，将文化认同、精神联系和生态服务统一于一体。这一案例对中国海洋治理中的社区参与、渔民传统知识整合以及海洋文化遗产保护具有独特启示价值。

☐ DOI: <https://doi.org/10.1038/s43016-026-01324-1>

133. Plant Community Responses to Long-Term Nutrient Additions Interact With Elevation and Vegetation Type in a Subarctic Tundra

【亚北极苔原植物群落对长期养分添加的响应与海拔和植被类型交互作用】

☐ Journal of Vegetation Science | Wed, 04 Mar 2026 21: 【Latest】 | 重要性评分: 38

☐ Abstract:

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ 中文摘要:

本研究在亚北极苔原开展长期养分添加实验，发现植物群落对氮磷添加的响应受海拔梯度和植被类型的强烈调控，不同海拔和植被类型下群落组成的变化方向和幅度存在显著差异。

□ 深度解读:

本研究发表于 *Journal of Vegetation Science*，通过长期实验揭示了苔原植被对养分富集响应的空间异质性。在全球变暖加速氮沉降和冻土融化释放养分的背景下，苔原植被的响应并非均质的——低海拔灌木苔原对养分更敏感，而高海拔地衣苔原则更抗干扰。这种差异化响应对预测北极植被绿化趋势和碳循环变化具有重要意义。

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70125>

134. Functional Traits Mediate Physico-Chemical Niche Differentiation of Epiphytic Lichens and Bryophytes in Temperate Forests

【功能性状介导附生地衣的理化生态位分化】

□ *Journal of Vegetation Science* | Tue, 17 Mar 2026 21: 【Latest】 | 重要性评分: 36

□ Abstract:

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

□ 中文摘要:

本研究揭示附生地衣通过功能性状（叶状体形态、光合色素、含水量等）的差异化实现理化微环境的生态位分化，解释了树干微尺度上多种地衣物种共存的机制。

□ 深度解读:

本研究发表于 *Journal of Vegetation Science*，将功能生态学方法应用于附生地衣群落研究，揭示了这些看似‘随机’分布在树干上的隐花植物实际通过精细的功能性状差异化占据不同的微生态位（光照、湿度、pH 梯度）。地衣是空气质量和生态系统完整性的经典指示生物，对城市生态学和森林健康监测具有应用价值。

□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70127>

135. Effects of Ungulate Herbivores on Temperate Forest Understory Vegetation—Implications From a Large-Scale Wildlife Exclosure Experiment in Central Europe

【有蹄类食草动物对温带森林林下植被的影响——对森林韧性的启示】

□ *Journal of Vegetation Science* | Sun, 08 Mar 2026 21: 【Latest】 | 重要性评分: 36

□ Abstract:

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

□ 中文摘要:

本研究综合分析了有蹄类食草动物（鹿、野猪等）对温带森林林下植被组成、结构和更新的影响，探讨这些影响对森林生态系统韧性（应对干扰后恢复的能力）的深远意义。

□ 深度解读:

本研究发表于 *Journal of Vegetation Science*，系统评估了有蹄类过度啃食对温带森林更新能力的威胁。过度的鹿群密度可能导致林下幼苗稀缺、灌木层消失和物种组成简化，从根本上削弱森林面对气候变化、病虫害等干扰后的恢复能力。这对中国东北和西南天然林保护区中鹿群管理和森林更新策略具有实践指导意义。

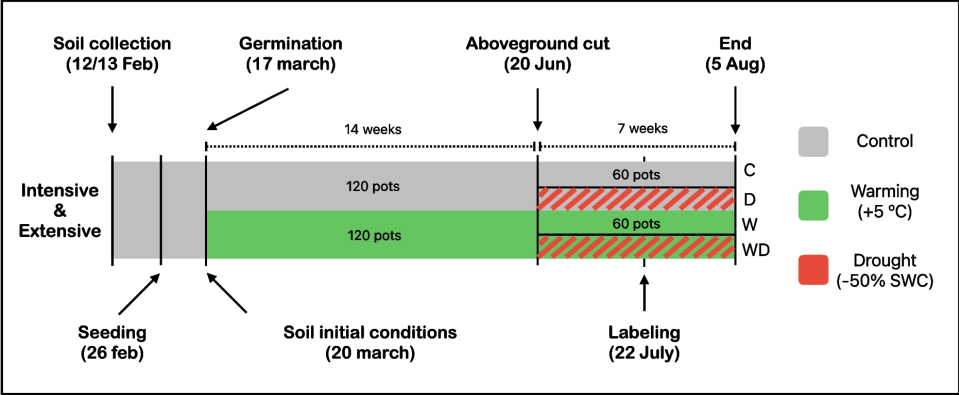
□ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70128>

136. Drought and warming enhance the risk of nitrogen losses in intensively managed grasslands

【干旱和增温加剧集约化管理草地的氮损失风险】

□ Plant and Soil | 2026-03-20 【Latest】 | 重要性评分: 35

□ Graphical Abstract:



□ Abstract:

Background and aims Climate change is expected to intensify drought and heatwaves, with major consequences for nutrient cycling in grasslands. Plant-soil-microbe interactions regulate nitrogen (N) dynamics, yet their responses to simultaneous warming and drought remain unclear, especially across plant species and land management histories. Methods We conducted a factorial warming and drought manipulation on six temperate grassland species belonging to three functional groups (grasses, forbs, legumes) grown in soils from intensively (Int) and extensively (Ext) managed grasslands. We quantified N content and biomass production across soil, microbes, roots, and shoots. We inferred fungi:bacteria by microbial C:N, not by direct soil community determination. Results Under control and warming co

□ 中文摘要:

本研究揭示干旱和气候变暖共同作用下，集约化管理草地的氮损失风险显著增加。干旱-复水循环和升温加速了氮的矿化和流失过程，可能导致地下水硝酸盐污染加剧和温室气体 N₂O 排放增加。

□ 深度解读:

本研究发表于 Plant and Soil，揭示了气候变化与农业管理交互作用下的氮循环风险。集约化草地大量施氮，而气候变暖和干旱事件加速了氮的转化和损失，导致‘施肥越多、损失越大’的恶性循环。这对中国北方牧区草地管理和化肥减量增效政策具有直接警示意义。

□ DOI: <https://link.springer.com/article/10.1007/s11104-026-08471-1>

137. Dual transcriptional control by ZmMYB127 regulates grain yield and quality

□ Nature Plants | 2026-03-19 【Latest】 | 重要性评分: 34

□ Abstract:

Nature Plants, Published online: 19 March 2026; doi:10.1038/s41477-026-02241-8 We identify ZmMYB127 as a central regulator of maize endosperm filling, which executes dual transcriptional activities to coordinate development of the aleurone layer and nutrient accumulation in grain. This coordination enhances grain yield and quality, underscoring ZmMYB127 as a potential target for improving maize breeding.

□ DOI: <https://doi.org/10.1038/s41477-026-02241-8>

138. Stress drives plasticity in leaf ageing transcriptional dynamics in *Arabidopsis thaliana*

☐ Nature Plants | 2026-03-19 **【Latest】** | 重要性评分: 34

☐ Abstract:

Nature Plants, Published online: 19 March 2026; doi:10.1038/s41477-026-02254-3 An atlas of ~1 million *Arabidopsis* nuclei captures leaf development under drought stress. Drought advances leaf maturation transcriptional programmes in a dose-dependent manner. Mesophyll-specific FRO6 overexpression partially restores growth, revealing a link between stress and development.

☐ DOI: <https://doi.org/10.1038/s41477-026-02254-3>

139. ZmMYB127 controls maize endosperm filling via dual-transcriptional regulation to improve grain yield and quality

☐ Nature Plants | 2026-03-19 **【Latest】** | 重要性评分: 34

☐ Abstract:

Nature Plants, Published online: 19 March 2026; doi:10.1038/s41477-026-02238-3 ZmMYB127 is a key regulator of maize endosperm filling, executing dual transcriptional control to coordinate aleurone development and nutrient accumulation. This coordination enhances grain yield and quality, underscoring its breeding potential.

☐ DOI: <https://doi.org/10.1038/s41477-026-02238-3>

140. A nitrogen-response coordinator

☐ Nature Plants | 2026-03-18 **【Latest】** | 重要性评分: 34

☐ Abstract:

Nature Plants, Published online: 18 March 2026; doi:10.1038/s41477-026-02265-0 A nitrogen-response coordinator

☐ DOI: <https://doi.org/10.1038/s41477-026-02265-0>

141. A systematic review of sustainable food systems identifies socio-economic pathways driving food systems transformations

☐ Nature Food | 2026-03-16 **【Latest】** | 重要性评分: 34

☐ Abstract:

Nature Food, Published online: 16 March 2026; doi:10.1038/s43016-026-01317-0 Studies on food systems transformations are heterogeneous. This systematic review develops pathways aimed at six key food systems actors across socio-economic contexts that can shift food systems towards greater sustainability.

☐ DOI: <https://doi.org/10.1038/s43016-026-01317-0>

142. Soil Metabarcoding Helps Identify Recalcitrant Taxa From Chaparral Seed Banks

☐ Journal of Vegetation Science | Wed, 18 Mar 2026 07: **【Latest】** | 重要性评分: 34

☐ Abstract:

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70124>

143. TaHST2 silencing shapes basal heat tolerance in allohexaploid wheat

☐ Nature Plants | 2026-03-20 **【Latest】** | 重要性评分: 32

☐ Abstract:

Nature Plants, Published online: 20 March 2026; doi:10.1038/s41477-026-02257-0 TaHST2 negatively regulates thermotolerance in wheat as it encodes a ubiquitin hydrolase that represses heat tolerance by stabilizing TaHSC701/702. During domestication, TaHST2 was progressively suppressed to select for improved thermotolerance.

☐ DOI: <https://doi.org/10.1038/s41477-026-02257-0>

144. The socioeconomic conditions that drive transformation

☐ Nature Food | 2026-03-16 **【Latest】** | 重要性评分: 32

☐ Abstract:

Nature Food, Published online: 16 March 2026; doi:10.1038/s43016-026-01320-5 Food systems encompass broad socioeconomic contexts involving several actors and sectors of society. Now, seven types of transition and six socioeconomic drivers have been identified that can facilitate or hinder sustainable food system transformations.

☐ DOI: <https://doi.org/10.1038/s43016-026-01320-5>

145. The Apple Does Not Fall Far From the Tree and How This Affects the Abiotic Stress Tolerance in Juvenile Trees

☐ Journal of Vegetation Science | Sat, 07 Mar 2026 21: **【Latest】** | 重要性评分: 32

☐ Abstract:

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ DOI: <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70126>

146. Cost-effectiveness of food fortification

☐ Nature Food | 2026-03-18 **【Latest】** | 重要性评分: 30

☐ Abstract:

Nature Food, Published online: 18 March 2026; doi:10.1038/s43016-026-01334-z Cost-effectiveness of food fortification

☐ DOI: <https://doi.org/10.1038/s43016-026-01334-z>

147. Stand development drives variation in the rhizosphere soil nitrogen-fixing community and nitrogen fractions in Biancaea sappan plantations in karst rocky desertification areas

☐ Plant and Soil | 2026-03-19 **【Latest】** | 重要性评分: 22

☐ Abstract:

Background and aims In karst rocky desertification areas, vegetation restoration is frequently limited by nitrogen availability. Although afforestation modifies soil nitrogen dynamics and nitrogen-fixing microbial communities, the relationships between soil nitrogen fractions and these microbial shifts remain unclear. This study investigated Biancaea sappan plantations across different age classes to examine how stand age influences nitrogen-fixing

microorganisms in the rhizosphere soil and their links to soil nitrogen fractions. **Methods** We analyzed soil chemical properties and nitrogen-fixing microbial characteristics across forest age classes in rocky desertification areas, examining changes in microbial diversity and soil nitrogen dynamics with stand age. **Results** Soil nitrogen fractions

□ DOI: <https://link.springer.com/article/10.1007/s11104-026-08486-8>

148. Independence of fine-root anatomical traits and their responses to long-term nitrogen addition in a temperate forest

□ Plant and Soil | 2026-03-19 **【Latest】** | 重要性评分: 21

□ Abstract:

Background and aims Long-term nitrogen (N) enrichment can substantially alter fine-root traits, which are central to predicting plant adaptation under global change. However, the multidimensional nature of fine-root morphology, chemistry, and anatomy—and the coordination among these trait dimensions—remains insufficiently understood. We aimed to assess whether long-term nitrogen addition modifies the multidimensional organization of fine-root traits, with particular emphasis on the role of root anatomical structure in trait coordination. **Methods** We used a 15-year N addition experiment in a mature temperate broadleaf-Korean pine mixed forest and measured morphological, chemical, and anatomical traits of first- to third-order fine roots across three dominant tree species (*Pinus koraiensis*, *A*

□ DOI: <https://link.springer.com/article/10.1007/s11104-026-08488-6>

149. Co-designing a research agenda for UK agroforestry using a multi-actor approach

□ Agronomy for Sustainable Development | 2026-02-27 **【Latest】** | 重要性评分: 19

□ Abstract:

There is growing recognition of agroforestry's potential to help mitigate and provide resilience to the climate and biodiversity crises. Beyond its environmental benefits, agroforestry can also enhance production and profits, making it a sustainable farming solution that is scalable. Despite this, uptake within Europe is low, and many knowledge gaps remain that need to be addressed to promote adoption and optimize the management and implementation of agroforestry systems. We co-developed a research agenda for agroforestry using a multi-actor approach and a modified Delphi method in 2023. 156 UK-based stakeholders contributed to this process, including farmers, advisors, policy makers, NGOs, and researchers. An initial list of 238 research priorities (high-priority research questions) was s

□ DOI: <https://link.springer.com/article/10.1007/s13593-026-01089-8>

150. Optimizing straw return management to suppress plant disease and promote plant growth. A meta-analysis

□ Agronomy for Sustainable Development | 2026-03-03 **【Latest】** | 重要性评分: 18

□ Abstract:

Straw return improves soil and nutrient cycling but may aggravate diseases if mismanaged. However, the comprehensive impact of different straw return methods on plant diseases remains unclear. This study aims to clarify the effects of straw return on plant diseases and explore appropriate straw return methods. This study evaluated the effect of straw return on the reduction of plant diseases by summarizing 1352 pairs of observations from 165 peer-reviewed publications. It further analyzed the effects of straw return conditions, disease types, and plant species on plant diseases. Overall, the study's findings showed that, with significant variation among return treatments, straw return reduced plant disease severity by 40.55% and increased plant biomass by 29.05%. Direct straw return of dif

□ DOI: <https://link.springer.com/article/10.1007/s13593-026-01094-x>

151. FarmSTEPS, a model for spatial-temporal exploration of diversified and sustainable crop sequence plans for crop farms

☐ Agronomy for Sustainable Development | 2026-03-17 【Latest】 | 重要性评分: 16

☐ Abstract:

Crop diversification could enhance economic and environmental sustainability of crop farming, but achieving this requires careful planning that accounts for crop interrelationships in a rotation or spatial layout. Several models have been devised to design diversified crop rotations or crop spatial layouts, but they often fail to capture farm settings, production constraints, and sustainability objectives specific to a farm at both temporal and spatial scales. To fill this gap, we developed a novel rule-based whole-farm planning model, FarmSTEPS, for systematically exploring diversified and sustainable farm crop sequence plans in a spatial-temporally explicit manner. FarmSTEPS comprises multiple farm characterization modules, an algorithm for generating farm crop sequence plans, and a mult

☐ DOI: <https://link.springer.com/article/10.1007/s13593-026-01097-8>

152. Precision planting of cereal rye cover crop improves sweet corn yield and farm benefits

☐ Agronomy for Sustainable Development | 2026-03-17 【Latest】 | 重要性评分: 14

☐ Abstract:

Including cereal rye (*Secale cereale* L.) in sweet corn (*Zea mays* L.) production systems provides environmental and soil benefits. However, cereal rye often decreases sweet corn yield and thus farm profit. A novel approach to reduce this negative effect is precision planting of cereal rye, which skips the sweet corn row to minimize crop interactions, potentially increasing nitrogen (N) availability, and reducing negative effects of cereal rye on sweet corn. A four-site-yr trial was conducted during the 2022-2024 growing seasons, the first known evaluation of precision-planted cereal rye compared to a no-cover crop control and solid-planted cereal rye. The study evaluated (i) cereal rye morphology, biomass production and its quality, (ii) sweet corn plant population, morphology, and marketab

☐ DOI: <https://link.springer.com/article/10.1007/s13593-026-01095-w>

153. Enhancing soil biological quality by increasing soil organic matter and microarthropod richness in semi-arid agricultural ecosystems: Fresh surpasses dried plant clipping amendments

☐ Agriculture Ecosystems & Environment 【Latest】 | 重要性评分: 12

☐ Abstract:

Publication date: 1 July 2026Source: Agriculture, Ecosystems & Environment, Volume 404Author(s): Huimin Shao, Zhen Cheng, Dong Wang, Jiangbo Qiao, Yingge Xie, Yuanjun Zhu, Gao-Lin Wu

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0167880926001647>

154. Molecular mechanisms of heavy metal uptake, accumulation, and tolerance in sweet potato (*Ipomoea batatas*): implications for phytoremediation and food safety

☐ Plant and Soil | 2026-03-20 【Latest】 | 重要性评分: 10

☐ Abstract:

Background and aims Soil contamination with heavy metals (HMs) is a major threat to agricultural productivity, ecosystem health, and food safety. Sweet potato (*Ipomoea batatas*) accumulates HMs in roots and tubers, affecting crop quality and its phytoremediation potential. This review summarizes molecular and physiological mechanisms of HM uptake, transport, sequestration, detoxification, and tolerance. It also highlights research gaps, particularly in

transporter kinetics and cultivar-specific tolerance, and suggests priorities for future work to improve phytoremediation and food safety. Results This review integrates knowledge on HM uptake pathways in sweet potato, including apoplastic and symplastic transport. These processes are mediated by zinc/iron-regulated transporter-like proteins

□ DOI: <https://link.springer.com/article/10.1007/s11104-026-08487-7>

155. An integrated soil–plant-microbiome framework for nanobubble-assisted reclamation of saline-sodic soils, with *Arundo donax* as a bioindicator

□ Plant and Soil | 2026-03-19 【Latest】 | 重要性评分: 10

□ Abstract:

Background and Aims Soil salinization limits plant growth and soil biology through increased electrical conductivity (EC), sodium adsorption ratio (SAR), and alkalinity. We tested whether air micro-nanobubble (AMNB) irrigation can reduce salinity and alkalinity in artificial saline-alkali soil, improve *Arundo donax* performance, and whether these effects correspond with changes in the rhizosphere microbiome. Methods A pot experiment compared non-saline and saline-alkali soils irrigated with AMNB or deionized water. We measured soil EC, pH, SAR, inorganic N and key hydrolase and oxidoreductase activities, and quantified *Arundo donax* growth, leaf water status, PSII photochemistry, antioxidant capacity and ion concentrations in roots and leaves. Rhizosphere bacterial communities were profiled

□ DOI: <https://link.springer.com/article/10.1007/s11104-026-08472-0>

【土壤与生物地球化学】

156. Alternative dynamic regimes of marine biogeochemical models in perturbed environments

【扰动环境中海洋生物地球化学模型的替代动态状态】

□ Biogeosciences | Tue, 17 Mar 2026 18: 【Latest】 | 重要性评分: 59

□ Abstract:

Alternative dynamic regimes of marine biogeochemical models in perturbed environments Guido Occhipinti, Davide Valenti, and Paolo Lazzari Biogeosciences, 23, 2003–2021, <https://doi.org/10.5194/bg-23-2003-2026>, 2026 Due to climate change shifts in ecosystem structure and function have been increasingly documented in marine ecosystems around the globe. We tested whether a marine biogeochemical model can predict shifts to alternative regimes in plankton and biogeochemical processes under environmental perturbations. Simulations show that perturbations can drive the system into new regimes, with responses that are either reversible or hysteretic, depending on the type and intensity of the disturbance.

□ 中文摘要:

由于气候变化，生态系统结构和功能转变正在加速。本研究分析了扰动环境中海洋生物地球化学模型可能出现的替代稳定状态，揭示海洋营养级联响应气候变化可能存在非线性突变，即生态系统可能发生不可逆的状态跳跃。研究识别了触发状态转换的关键阈值（温度、酸化、富营养化），为理解海洋生态系统的临界点提供了理论框架。

□ 深度解读:

本研究发表于 Biogeosciences，通过数学分析和模型模拟，首次系统揭示海洋生物地球化学系统在气候变化压力下可能存在多个替代稳定状态。研究发现营养盐-浮游生物-溶解氧系统在不同扰动情景下可能发生突变式跳跃，且这种状态转换可能具有滞后效应——即使环境压力恢复，系统也不一定回到原始状态。这对预测未来海洋生产力变化和碳泵效率具有重要意义，对中国黄海、东海等近海富营养化海域的长期管理决策具有直接参考价值。

□ DOI: <https://doi.org/10.5194/bg-23-2003-2026>

157. Hydraulic Redistribution Decreases with Precipitation Magnitude and Frequency in a Dryland Ecosystem: A Data-Model Fusion Approach

【旱地生态系统中水力再分配随降水量和频率增加而减弱：数据-模型融合方法】

□ Biogeosciences | Wed, 18 Mar 2026 18: 【Latest】 | 重要性评分: 42

□ Abstract:

Hydraulic Redistribution Decreases with Precipitation Magnitude and Frequency in a Dryland Ecosystem: A Data-Model Fusion Approach Aneesh Kumar Chandel, Mitra Cattray, Yu Zhou, Hang Duong, Marcy E. Litvak, William T. Pockman, and Yiqi Luo Biogeosciences, 23, 2045–2058, <https://doi.org/10.5194/bg-23-2045-2026>, 2026 Hydraulic redistribution (HR) is a passive process in which water can move between wet and dry regions in the root zone by flowing through plant root systems. In this modeling study, we showed that adding HR to a process-based model improved soil moisture predictions, particularly in the top 30 cm. HR rates declined with increasing rainfall magnitude and frequency, but HR rates were also influenced by the length of dry spells between rainfall events.

□ 中文摘要:

本研究利用数据-模型融合方法，发现旱地生态系统中植物根系的水力再分配（通过深根将深层水分输送至浅层干燥土壤的过程）强度随降水量和降水频率增加而减弱。这一发现对理解旱地生态系统的水分循环和植被适应策略具有重要意义。

□ 深度解读:

本研究发表于 Biogeosciences，利用先进的数据同化技术，首次定量揭示了降水格局对旱地植物水力再分配的调控规律。水力再分配是旱地生态系统中一种关键的‘地下灌溉’机制，深根植物通过根系网络将深层水分被动输送至浅层土壤，惠及浅根邻居物种。研究发现随着降水增加该机制弱化，意味着在气候变化导致的极端干旱事件中，水力再分配可能更加重要。对中国西北干旱半干旱区生态恢复策略具有参考价值。

□ DOI: <https://doi.org/10.5194/bg-23-2045-2026>

158. Carbon soil stock change in an intensive crop field near Paris reveals significant carbon losses over a decade

【巴黎近郊集约化农田十年间揭示显著土壤碳损失】

□ Biogeosciences | Wed, 18 Mar 2026 18: 【Latest】 | 重要性评分: 38

□ Abstract:

Carbon soil stock change in an intensive crop field near Paris reveals significant carbon losses over a decade Benjamin Loubet, Nicolas P. A. Saby, Bruna Winck, Maryam Gebleh, Pauline Buysse, Jean-Philippe Chenu, Céline Ratié, Claudy Jolivet, Carmen Kalalian, Florent Levavasseur, Jose-Luis Munera-Echeverri, Sébastien Lafont, Denis Loustau, Dario Papale, Giacomo Nicolini, and Dominique Arrouays Biogeosciences, 23, 2059–2077, <https://doi.org/10.5194/bg-23-2059-2026>, 2026 Soil is a large pool of carbon storing globally, from 2 to 3 times more carbon than the atmosphere and vegetation. We measure the soil stock change from 2005 to 2019 at the ICOS crop site FR-Gri near Paris. The soil carbon stock decreased by approximately $72 \pm 17 \text{ g C m}^{-2} \text{ yr}^{-1}$ over the period, resulting in a total loss of aro

□ 中文摘要:

本研究对巴黎近郊一块集约化作物农田进行了十年期的土壤碳储量变化监测，发现即使在常规管理条件下，集约化农田也存在显著的土壤有机碳净损失，对欧洲农业碳中和目标构成挑战。

□ 深度解读:

本研究发表于 Biogeosciences，通过十年连续监测提供了集约化农业导致土壤碳持续流失的直接证据。研究所在的巴黎盆地是欧洲最重要的粮食产区之一，该区域土壤碳的净损失对欧洲农业碳中和目标（Farm to Fork 策略）构成严峻挑战。这一长期数据对中国华北平原等类似集约化农区的土壤碳动态研究和碳汇核算具有重要对比参考价值。

□ DOI: <https://doi.org/10.5194/bg-23-2059-2026>

159. The 3D submicron-scale skeletal reconstruction of Nannoconus (Cretaceous calcareous nannofossil) –Insights into biomineralization

☐ Biogeosciences | Wed, 18 Mar 2026 18: **【Latest】** | 重要性评分: 36

☐ Abstract:

The 3D submicron-scale skeletal reconstruction of Nannoconus (Cretaceous calcareous nannofossil) – Insights into biomineralization Rajkumar Chowdhury, Redhouane Boudjehem, Baptiste Suchéras-Marx, Maxime Dupraz, Anico Kulow, Julio Cesar da Silva, Jean Louis Hazemann, Marie-Pierre Aubry, Javier Pérez, Alejandro Fernandez-Martinez, and Fabienne Giraud Biogeosciences, 23, 2023–2043, <https://doi.org/10.5194/bg-23-2023-2026>, 2026 Nannoconus, a major marine planktonic biocarbonate producer of the Early Cretaceous (150–120 Ma), biomineralized heavy (200–1400 picogram) skeletons (5–20 µm) of interlocking lamellae spanned around a central canal, by an unknown process. With the first-ever application of synchrotron-based ptychographic X-ray computed tomography (PXCT) of nanometric resolution, we reco

☐ DOI: <https://doi.org/10.5194/bg-23-2023-2026>

160. The Farmer and Soil Health: Understanding the Foundation for Sustainable Soil Management

☐ European Journal of Soil Science | Sun, 15 Mar 2026 21: **【Latest】** | 重要性评分: 34

☐ Abstract:

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70294>

161. The Role of Autonomous Mechanical Weeding Robots in Climate-Smart Soil Management: A Scoping Review

☐ European Journal of Soil Science | Wed, 11 Mar 2026 04: **【Latest】** | 重要性评分: 34

☐ Abstract:

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70302>

162. Trade-Off Analysis Between Environmental Effects and Profitability in Agriculture: A Finnish Case-Study

☐ European Journal of Soil Science | Wed, 11 Mar 2026 00: **【Latest】** | 重要性评分: 34

☐ Abstract:

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70300>

163. Soil Management Effects on Grapevine Water Uptake Depth in a Mediterranean Vineyard

☐ European Journal of Soil Science | Mon, 09 Mar 2026 00: **【Latest】** | 重要性评分: 34

☐ Abstract:

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70301>

164. The distribution and isotopomeric characterization of nitrous oxide in the Eastern Gotland Basin (central Baltic Sea)

☐ Biogeosciences | Tue, 17 Mar 2026 18: **【Latest】** | 重要性评分: 32

☐ Abstract:

The distribution and isotopomeric characterization of nitrous oxide in the Eastern Gotland Basin (central Baltic Sea) Pratihara Bardhan, Claudia Frey, Gregor Rehder, and Hermann W. Bange Biogeosciences, 23, 1987–2002, <https://doi.org/10.5194/bg-23-1987-2026>, 2026 Nitrous oxide (N₂O), a potent greenhouse gas, is released from coastal seas & estuaries, yet we don't fully understand how it is formed and consumed. In this study we collected water from several sites in the central Baltic Sea. N₂O came from ammonia in oxic waters. Deep waters with low to no oxygen noted more active N₂O cycling. The seafloor was a source in some areas. Typically N₂O is produced by bacteria, but our results indicate possibility of other players like fungi or chemical reactions.

☐ DOI: <https://doi.org/10.5194/bg-23-1987-2026>

165. Evaluating Soil Aeration Dynamics in a Non-Rigid Vertisol Through Shrink and Swelling Curves During Wetting-Drying Cycle

☐ European Journal of Soil Science | Sun, 15 Mar 2026 16: **【Latest】** | 重要性评分: 32

☐ Abstract:

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ DOI: <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70304>

166. Nitrate sources and transformation pathways in salt-affected agricultural ditches: biogeochemical responses along contrasting salinity gradients

☐ Biogeochemistry | 2026-03-09 **【Latest】** | 重要性评分: 18

☐ Abstract:

Agricultural nitrate pollution, largely associated with high-intensity fertilizer application and untreated or insufficiently treated waste inputs, has become a major driver of water quality degradation worldwide. Agricultural ditches, as key recipients of surface runoff and lateral seepage, function as both transport conduits and biogeochemical hotspots for nitrogen transformation. In salt-affected ditches, high nitrogen inputs and salinity can interact to enhance nitrate accumulation and mobility, thereby increasing the potential for downstream transport. To bridge this knowledge gap, we investigated two representative saline agricultural ditches located in northwestern China, a region characterized by an arid climate and saline–alkaline soils. We employed stable isotope analysis (δ¹⁵N-N

☐ DOI: <https://link.springer.com/article/10.1007/s10533-026-01318-y>

167. Mobilized dissolved organic matter provides niche space for prokaryotes in the deep subterranean estuary of a sandy beach

☐ Biogeochemistry | 2026-03-17 **【Latest】** | 重要性评分: 16

☐ Abstract:

A fraction of organic carbon in beach sands is mobilizable as dissolved organic matter (DOM). The molecular composition of this mobilizable DOM and its relation to microbial metabolism is central for biogeochemical processes in beach ecosystems, yet, still poorly understood. To identify these DOM-microbe interactions, we analyzed two continuous 24-m-long sediment cores from the high-energy beach of Spiekeroog Island, Germany. The beach is characterized by a deep subterranean estuary (STE), where sediments are flushed by fresh, saline and brackish water. Even though beach sands are generally low in organic carbon, we found that concentrations of mobilizable

DOM from sediments were approximately 16 times higher than the in-situ groundwater concentrations. Ultrahigh-resolution mass spectrometry

☐ DOI: <https://link.springer.com/article/10.1007/s10533-026-01315-1>

168. No indications for a priming effect on soil organic carbon mineralization in a temperate river system

☐ Biogeochemistry | 2026-03-07 **【Latest】** | 重要性评分: 16

☐ Abstract:

Globally, rivers are vital conduits transporting and processing terrestrial carbon, and are generally considered to act as source of carbon dioxide (CO₂) towards the atmosphere. A large amount of soil organic carbon (SOC) is transferred from the land surface to river systems each year, where it mixes and interacts with the autochthonous carbon pool (i.e., produced in-stream via photosynthesis). The latter has been suggested to be more labile and to potentially affect—positively or negatively—the mineralization rate of the more recalcitrant SOC, a mechanism referred to as the priming effect (PE). Here, we performed series of short-term (7 days) incubation experiments to investigate whether the addition of (¹³C-labelled) algal carbon (C) affected SOC mineralization in an aquatic environment,

☐ DOI: <https://link.springer.com/article/10.1007/s10533-026-01317-z>

169. Soil heterotrophic respiration after irrigation retirement is differentially influenced by moisture and substrate availability over time

☐ Biogeochemistry | 2026-03-09 **【Latest】** | 重要性评分: 14

☐ Abstract:

Water limitations are forcing producers to transition large areas of currently irrigated farmland into dryland agriculture across the Western U.S. with unclear effects on global soil carbon (C) dynamics. An experiment established in 2017 in a no-till, maize system in Colorado suggested that soil heterotrophic respiration (Rh) following irrigation retirement was co-regulated by water and available C. We continued Rh measurements in 2021–2022 along with monthly soil samplings to explore the interactive effects of soil moisture and available C on microbial community composition and activity. Plant C inputs, available soil water, bacteria, fungi, and protozoa fatty acid methyl ester (FAME) biomarkers, enzyme activity, and Rh decreased after irrigation retirement, while actinobacteria abundance

☐ DOI: <https://link.springer.com/article/10.1007/s10533-026-01312-4>

170. Precipitation legacy effects on gross N mineralization and nitrification rates in a Pinyon-Juniper dryland under altered precipitation

☐ Biogeochemistry | 2026-03-09 **【Latest】** | 重要性评分: 14

☐ Abstract:

Changes in precipitation patterns can influence soil nitrogen (N) cycling by altering the timing and extent of moisture availability. Arid conditions, associated with more frequent and severe droughts, can increase soil inorganic N concentrations when diffusion constraints decouple N sources and sinks, making this N susceptible to loss. Further, shifts in precipitation patterns in one season can “carry over” N into subsequent seasons, known as precipitation legacies. However, how these legacies increase soil N availability through changes in inorganic N production (i.e., gross N mineralization and nitrification) remain unclear. We manipulated the amount and timing of precipitation in a pinyon–juniper dryland by either adding or excluding precipitation during the winter or summer months and

☐ DOI: <https://link.springer.com/article/10.1007/s10533-026-01313-3>

171. Cover crops shape soil microbiome and ecological interactions in the soybean rhizosphere under long-term no-till

☐ Soil & Tillage Research 【Latest】 | 重要性评分: 11

☐ Abstract:

Publication date: September 2026
Source: Soil and Tillage Research, Volume 261
Author(s): Caio César Gomes Freitas, Carolinne Rosa de Carvalho, Nariane de Andrade, Rafael Lima Oliveira, Victor Lucas Vieira Prudêncio Araújo, Felipe Martins do Rêgo Barros, João Henrique dos Santos Ferreira, José Roberto Portugal, Carlos Alexandre Costa Crusciol, Fernando Dini Andreote

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S016719872600108X>

172. Long-term N fertilization altered viral community composition and increased functional potentials of phosphorus cycling by causing soil acidification

☐ Soil Biology and Biochemistry 【Latest】 | 重要性评分: 9

☐ Abstract:

Publication date: June 2026
Source: Soil Biology and Biochemistry, Volume 217
Author(s): Shuxun Cheng, Zhongmin Dai, Dan He, Ran Xue, Kankan Zhao, Jorge L. Mazza Rodrigues, Lucas W. Mendes, Scott X. Chang, Jianming Xu

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0038071726000568>

【遥感与模型】

173. Quantifying Potential Nutrient Liberation Following Plant Investment in Root Exudates: Variation Across Soil Profiles and Ecosystems in the United States

【量化植物根系分泌物投资后的潜在养分释放：美国不同土壤剖面 and 生态系统的差异】

☐ Journal of Geophysical Research - Biogeosciences | Fri, 13 Mar 2026 04: 【Latest】 | 重要性评分: 38

☐ Abstract:

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ 中文摘要:

本研究量化了植物根系分泌物对土壤养分（氮、磷）释放的促进作用，分析了这一过程在美国不同生态系统和土壤深度间的变异规律，揭示了根际过程在土壤养分循环中的关键调控作用。

☐ 深度解读:

本研究发表于 Journal of Geophysical Research - Biogeosciences，系统量化了一个长期被低估的地下碳-养分循环耦合机制：植物通过向土壤释放有机酸等根系分泌物，‘投资’碳换取矿物结合态养分的释放。研究发现这一机制在养分贫瘠的深层土壤中尤为重要，可解释深根植物在养分匮乏环境中的竞争优势。对理解森林和草地生态系统的养分获取策略具有理论意义。

☐ DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009380>

174. Nonlinear Elevation-Dependent Controls on Drought Responses of Vegetation Net Primary Production in the Central Himalayas

☐ Journal of Geophysical Research - Biogeosciences | Tue, 17 Mar 2026 06: 【Latest】 | 重要性评分: 34

☐ Abstract:

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009490>

175. Contrasting CO₂ Fluxes Among Arctic Pond Types Are Driven by Differences in Carbon Lability, Terrestrial Inputs, and Benthic Respiration

☐ Journal of Geophysical Research - Biogeosciences | Thu, 12 Mar 2026 06: **【Latest】** | 重要性评分: 32

☐ Abstract:

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009318>

176. Gold-Standard Science in Uncertain Times —Thank You to the 2025 Reviewers of JGR Biogeosciences

☐ Journal of Geophysical Research - Biogeosciences | Sat, 14 Mar 2026 02: **【Latest】** | 重要性评分: 30

☐ Abstract:

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2026JG009838>

177. Influence of Biofilms and Extracellular Polymeric Substances on High-Frequency Acoustic Backscatter

☐ Journal of Geophysical Research - Biogeosciences | Thu, 12 Mar 2026 06: **【Latest】** | 重要性评分: 30

☐ Abstract:

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 3, March 2026.

☐ DOI: <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009446>

178. A novel soil moisture retrieval method via combining radiative transfer model and machine learning

☐ Remote Sensing of Environment **【Latest】** | 重要性评分: 18

☐ Abstract:

Publication date: 15 May 2026
Source: Remote Sensing of Environment, Volume 338
Author(s): Yurun Chen, Cheng Tong, Josh Qixuan Sun, Yulin Shangguan, Xiaodong Deng, Mark Crowley, Hongquan Wang, Yang Ye, Haijun Bao, Ruqi Huang

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001483>

179. Canopy water content is related to variability in leaf water over time but not across space in an oak savanna ecosystem

☐ Remote Sensing of Environment **【Latest】** | 重要性评分: 13

☐ Abstract:

Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Jean Allen, Indra Boving, Leander D.L. Anderegg, Philip G. Brodrick, K. Dana Chadwick, Dar Roberts, Anna T. Trugman

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001446>

180. Pulse fragmentation-induced uncertainty in forest LAI mapping using UAV LiDAR

☐ Remote Sensing of Environment **【Latest】** | 重要性评分: 11

☐ Abstract:

Publication date: 15 May 2026Source: Remote Sensing of Environment, Volume 338Author(s): Yuchen Bai, Philippe Verley, Tiangang Yin, Nicolas Lauret, Florence Forbes, Jean-Baptiste Durand, Grégoire Vincent

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001112>

181. Satellite-based reconstruction of high-resolution ocean subsurface temperature using spatiotemporal graph attention networks

☐ International Journal of Applied Earth Observation **【Latest】** | 重要性评分: 11

☐ Abstract:

Publication date: May 2026Source: International Journal of Applied Earth Observation and Geoinformation, Volume 149Author(s): Jiawei Xun, Jungang Yang, Shibao Li, Ole Baltazar Andersen, Wei Cui, Jie Sun, Haoran Gu

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S1569843226001536>

182. UAV-based deep learning for biodiversity monitoring: Advances, applications, and future directions

☐ Ecological Informatics **【Latest】** | 重要性评分: 11

☐ Abstract:

Publication date: May 2026Source: Ecological Informatics, Volume 95Author(s): Sheng Wang, Kexin Chen, Ziheng Wei, Linzhe Yang, Qiao Wang, Mingzhi Liu, Kangjian Cao, Chaoyue Zhao, Runlong Chang, Zhen Wang, Xuchen Wang, Meng Yang, Qiang Ji, Guodong Sun, Fu Xu

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S1574954126001160>

183. CE-CNN-FA-BiGRU: A novel deep learning framework for sap flow prediction across trees at different ages and species with historical environmental factors

☐ Ecological Informatics **【Latest】** | 重要性评分: 11

☐ Abstract:

Publication date: May 2026Source: Ecological Informatics, Volume 95Author(s): Yane Li, Yuhang Jiang, Lijun Guo, Weibo Wang, Jiahao Wu, Xiang Weng, Hailin Feng

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S1574954126001214>

184. Urban PolSAR decomposition: Challenges and new directions for emerging SAR technologies

☐ Remote Sensing of Environment **【Latest】** | 重要性评分: 9

☐ Abstract:

Publication date: 15 May 2026 Source: Remote Sensing of Environment, Volume 338 Author(s): Dingfeng Duan, Yong Wang

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001331>

185. National-scale mapping of plastic-mulched farmlands in China based on deep semantic segmentation and Sentinel-2 imagery

☐ Remote Sensing of Environment **【Latest】** | 重要性评分: 9

☐ Abstract:

Publication date: 15 May 2026 Source: Remote Sensing of Environment, Volume 338 Author(s): Bowen Niu, Quanlong Feng, Xinmin Zhang, Bingwen Qiu, Yelu Zeng, Shuai Su, Gaoyuan Xu, Jianhua Gong, Xiaolu Yan, Jianxi Huang, Gaofei Yin, Jiantao Liu, Jianyu Yang, Dehai Zhu

☐ DOI: <https://www.sciencedirect.com/science/article/pii/S0034425726001380>

【预印本】

186. Phenotypic plasticity evolved for climate variability constrains performance under climate warming

☐ bioRxiv Evolutionary Biology | 2026-03-20 **【Latest】** | 重要性评分: 27

☐ Abstract:

Phenotypic plasticity allows plants to rapidly respond to changing environments without the need for evolutionary change or migration. While selection can create variation in plasticity across natural populations, these responses are not adaptive in all environments. To predict whether plasticity will be adaptive requires evaluation of its fitness effects across a range of environments, including novel ones. Here, we test how traits and their plasticity vary for genotypes collected across a natural hybrid zone between two tree species with contrasting climatic niches. Fast-growing *Populus trichocarpa* inhabits maritime environments with relatively warm and stable temperatures, while *P. balsamifera* inhabits continental environments with cold winters and large temperature variance throughout

☐ DOI: <https://doi.org/10.1101/2026.03.15.711905>

187. Spectral Phenotyping Reveals Time-Specific QTLs in Field-Grown Lettuce

☐ bioRxiv Plant Biology | 2026-03-18 **【Latest】** | 重要性评分: 24

☐ Abstract:

Lettuce (*Lactuca sativa*) is an important field crop, but our understanding of its phenotypic variation and underlying genetics under natural field conditions remains limited, posing challenges for identifying effective crop breeding targets. Longitudinal hyperspectral phenotyping allows for non-invasive monitoring of crop performance under diverse agricultural conditions. In this study, we used hyperspectral imaging to assess the phenotypic variation of almost 200 different field-grown lettuce varieties, following the same plants from just after seedling- to flowering-stage. With automated image processing, we extracted a wide range of spectral phenotypes related to metabolite

content, growth efficiency, and environmental stress responses, creating a multi-dimensional time-resolved data se

☐ DOI: <https://doi.org/10.1101/2026.03.16.711173>

188. A holistic survey of small mammal diversity across an iconic Madrean Sky Island (Santa Catalina Mountains, Arizona, USA)

☐ bioRxiv Zoology | 2026-03-18 **【Latest】** | 重要性评分: 22

☐ Abstract:

The Santa Catalina Mountains are an iconic member of the Madrean Sky Islands, rising above Tucson, Arizona, USA, where the Catalina Highway connects Sonoran desertscrub to stands of conifer forest nearly 2,800 meters in elevation. As one of the [~]54 forested mountain areas in this system, the Santa Catalinas host unique biotic communities relative to the surrounding lowlands. However, most of these sky islands lack the surveys of resident small mammals (either historical or recent) needed for studying biodiversity in the context of changing climate and habitat use. From 2021 to 2023, we surveyed 10 localities on the north and south slopes of the Santa Catalina Mountains using holistic sampling methods to document terrestrial small mammal diversity and preserve multiple tissue types. Here

☐ DOI: <https://doi.org/10.1101/2026.03.15.711934>

189. Asymmetric depth acclimation and plasticity limit the refugial potential of mesophotic *Porites astreoides*

☐ bioRxiv Ecology | 2026-03-20 **【Latest】** | 重要性评分: 21

☐ Abstract:

Mesophotic coral ecosystems have been proposed as climate refugia for shallow reefs, yet the capacity of mesophotic corals to persist across depth gradients remains unresolved. We conducted a long-term reciprocal transplantation of the Caribbean coral *Porites astreoides* between shallow (10 m) and mesophotic (40 m) reefs to assess physiological, skeletal, and transcriptomic plasticity. Depth, rather than season, was the primary driver of coral performance. Shallow colonies exhibited higher metabolic activity and calcification, whereas mesophotic colonies showed reduced protein content, slower skeletal extension, and elevated expression of skeletal organic matrix genes. Transplant responses were asymmetric: shallow-to-deep corals acclimated through coordinated physiological and transcription

☐ DOI: <https://doi.org/10.1101/2026.03.18.712007>

190. Insights from whole-genome sequencing and comparative genomic analysis of the pathogenic *Photobacterium damsela* subsp. *damsela* strain MRY0520 isolated from *Litopenaeus vannamei*

☐ Research Square - Environment | 2026-03-20 21:11:24 **【Latest】** | 重要性评分: 19

☐ Abstract:

Photobacterium damsela is a widespread pathogenic bacterium in marine ecosystems, primarily associated with fish skin ulcers. However, infections caused by this bacterium in shrimp are rarely reported. The dominant strain MRY0520, isolated from the hepatopancreas of diseased *Litopenaeus vannamei*, was identified as *Photobacterium damsela* subsp. *damsela* (Phdd) and exhibited high pathogenicity. Elucidating the genetic architecture and infection mechanisms of *P. damsela* is crucial for the effective prevention and control of related diseases. To characterize its genomic features, whole-genome sequencing and comprehensive functional annotation were performed on the virulent Phdd strain MRY0520, followed by comparative genomic analysis to investigate its evolutionary traits. The genomic seque

☐ DOI: <https://www.researchsquare.com/article/rs-9004776/latest>

191. Resource abundance and dietary specialization predict elevational migration in a hyperdiverse montane bird community

☐ bioRxiv Ecology | 2026-03-20 **【Latest】** | 重要性评分: 18

☐ Abstract:

Migration is a well-described behavioural strategy that allows species to track variation in resources and climatic conditions by moving in response to seasonality. A common form is elevational migration, an annual short-distance movement undertaken by many mountain bird species globally. While studies show that the timing of migration may relate to food availability, the mechanisms determining which species migrate remain unclear. Our study investigated if the degree of dietary specialization explains why some high-elevation bird species in seasonal environments migrate downslope for the winter while others remain resident at high altitudes despite the apparent scarcity of their preferred food resources. We mist-netted birds along a 2300-m elevational gradient in the Eastern Himalaya and

☐ DOI: <https://doi.org/10.1101/2026.03.18.710293>

192. Failure to invest below-ground may limit the Northern expansion of invasive knotweed: lessons from a two-phase transplant experiment

☐ bioRxiv Ecology | 2026-03-20 **【Latest】** | 重要性评分: 18

☐ Abstract:

The ecological and evolutionary processes determining species range limits remain poorly understood. Ultimately, range limits depend on the species abilities to persist under heterogeneous conditions, by adaptive differentiation and phenotypic plasticity, including transgenerational effects. To investigate ecological differentiation and transgenerational effects in the clonal invasive knotweed, *Reynoutria japonica*, in Europe, we conducted a two-phase transplant experiment: plants sampled along the entire latitudinal gradient were planted in three sites located at the northern range margin, mid-range and near the southern range margin, and then re-transplanted among all three sites after two years. Biomass production and allocation were generally not associated with latitude of origin and p

☐ DOI: <https://doi.org/10.1101/2026.03.18.712549>

193. Gene Expansion and Regulatory Rewiring Shape Sex-Biased Evolution of the Mouse Submandibular Gland Secretome

☐ bioRxiv Evolutionary Biology | 2026-03-20 **【Latest】** | 重要性评分: 15

☐ Abstract:

Mammalian saliva plays essential roles in digestion, immunity, and host-microbiome interactions, yet its protein composition varies across species and sexes. The evolutionary mechanisms underlying this molecular diversity remain poorly understood. Here, we compared mouse and human salivary gland secretomes at genomic, transcriptomic, and proteomic levels to understand how saliva composition evolves. We performed RNA-seq analysis of the major mouse salivary glands (parotid, submandibular, and sublingual), liver and pancreas from both sexes, compared them with reanalyzed previously published human salivary gland transcriptomes, and integrated them with proteome data of mouse and human whole saliva. We found that evolution of gene expression in mouse salivary glands is driven by rapid gene tu

☐ DOI: <https://doi.org/10.1101/2026.03.18.712472>

194. TaxonMatch: taxonomic integration and tree construction from heterogeneous biological databases

☐ bioRxiv Evolutionary Biology | 2026-03-20 **【Latest】** | 重要性评分: 11

☐ Abstract:

Integrating taxonomic data from various sources presents a significant challenge in the study of biodiversity research, due to non-standardized nomenclature and evolving species classifications. Discrepancies between major repositories like the Global Biodiversity Information Facility (GBIF) and the National Center for Biotechnology Information (NCBI), as well as citizen science platforms such as iNaturalist, lead to fragmented and sometimes inaccurate biological data. We present TaxonMatch, a tool designed to address these challenges. TaxonMatch aligns taxonomic names, resolves synonymy, and corrects typographical and structural inconsistencies across databases. We show how it can be used to build a common backbone arthropod taxonomy over NCBI, GBIF and iNaturalist, to find the closest mo

☐ DOI: <https://doi.org/10.1101/2026.03.18.712418>

195. Weather Characterization for Optimizing Genomic Prediction in *Miscanthus sacchariflorus*

☐ bioRxiv Genomics | 2026-03-20 **【Latest】** | 重要性评分: 11

☐ Abstract:

Environmental factors affect crop growth and development thus their consideration across sites and years become essential for genotypic evaluation. Genomic selection (GS) has been broadly implemented to accelerate breeding cycles by skipping field evaluations thus allowing early identification of outperforming genotypes. In this study, 7,740 phenotypic records corresponding to 516 *Miscanthus sacchariflorus* genotypes evaluated in five locations across three years were considered for analysis. Additionally, environmental data on six weather covariates was implemented to characterize similarities between locations. Different sets of locations of variable sizes were used for model calibration based on two cross-validations (CV00 and CV0) schemes leaving out one location at a time. Predictive a

☐ DOI: <https://doi.org/10.1101/2026.03.18.712712>

196. OxBreaker: species-agnostic pipeline for the analysis of outbreaks using nanopore sequencing

☐ bioRxiv Genomics | 2026-03-19 **【Latest】** | 重要性评分: 11

☐ Abstract:

Real-time genomic surveillance may mitigate the increasing treatment costs of healthcare-associated infections, but whole-genome sequencing costs and the need for specialised expertise constrain its wide implementation in public health. Here we present 'OxBreaker', an automated and species-agnostic pipeline optimised for the high-resolution analysis of bacterial and plasmid genomes sequenced via Oxford Nanopore Technologies (ONT). 'OxBreaker' streamlines the transition from raw reads to phylogenetic inference through automated reference selection and high-accuracy variant calling. It is accessible via a graphical user interface (GUI) that can be easily installed locally and operated by non-specialists. Benchmarking against technical and biological replicates of high-priority pathogens demo

☐ DOI: <https://doi.org/10.1101/2026.03.18.709804>

197. Bed and breakfast in the bush: Selection of resting sites and kill sites by leopards (*Panthera pardus*) on Namibian farmland

☐ bioRxiv Ecology | 2026-03-20 **【Latest】** | 重要性评分: 10

☐ Abstract:

Sleeping and feeding are crucial for survival of any animal. In case of large predators, knowing where these activities occur can help us understand their behavioural adaptations for coexisting with people and could help mitigating human-carnivore conflicts. Leopard (*Panthera pardus*) is an elusive and highly adaptable large felid that mostly

lives outside protected areas and can survive also in close proximity to humans. However, most leopard research in Africa has been conducted in protected areas and we poorly understand habitat selection of leopards while resting and hunting. To shed light on their coexistence with humans, we investigated habitat features influencing leopard selection of resting and kill sites on farmlands in central Namibia, using generalized linear mixed models (GLMM)

☐ DOI: <https://doi.org/10.1101/2026.03.18.712594>

198. Introgression across ploidies contributes to genetic diversity in introduced urban *Capsella bursa-pastoris*

☐ bioRxiv Plant Biology | 2026-03-19 【Latest】 | 重要性评分: 10

☐ Abstract:

Successful establishment of a species in a new range is a useful way to understand the impact of demography and selection on the evolution of globally distributed species. In particular, introductions influence genetic diversity and population structure in the introduced range in unpredictable ways. Additionally, introgressive hybridization is often associated with successful establishment in new ranges. In this study, we explore the impact of introgressive hybridization on the polyploid *Capsella bursa-pastoris* in the New York City metropolitan area. We find *Capsella bursa-pastoris* in the New York City metropolitan area likely originated from multiple introductions from northern Eurasia, and that populations across the New York City metropolitan area are generally panmictic. As with *Capsella*

☐ DOI: <https://doi.org/10.1101/2026.03.17.712489>

199. Sowing date effects on anther dehiscence, pollen germination on the stigma, and fertility under heat in Japanese rice

☐ bioRxiv Plant Biology | 2026-03-19 【Latest】 | 重要性评分: 10

☐ Abstract:

Heat-tolerant rice cultivars are essential for mitigating global warming impacts. Basal anther dehiscence length (BDL) is a promising visible morphological marker for heat tolerance through stable pollination. We investigated the effects of sowing date on anther morphology, pollination, and fertility under controlled high-temperature conditions (35, 37, or 39 {degrees}C at flowering). Three japonica cultivars--'Akitakomachi' (early heading), 'Koshihikari' (medium), and 'Hatsushimo' (late)--were sown monthly over 3 months and grown in pots. At heading, the plants were exposed to the temperature treatments for 3 days, and the proportion of florets with ≥ 10 germinated pollen grains on the stigma (GP10) and seed set were assessed. Among anther traits, BDL showed the greatest variation, with

☐ DOI: <https://doi.org/10.1101/2026.03.17.712342>

200. Chromosome-scale genome of the woody oilseed crop *sacha inchi* elucidates the molecular basis of alpha-linolenic acid biosynthesis and triacylglycerol accumulation in seeds

☐ bioRxiv Genomics | 2026-03-20 【Latest】 | 重要性评分: 9

☐ Abstract:

Sacha inchi (*Plukenetia volubilis* L.) is an emerging woody oilseed crop prized for its high alpha-linolenic acid (ALA) content. Despite its nutritional and economic value, the lack of high-quality genomic resources has hindered genetic improvement and the elucidation of its unique polyunsaturated fatty acid and lipid biosynthetic pathways. In this study, we report a high-quality, chromosome-scale genome assembly of *sacha inchi* with a total length of 710.62 Mb, integrated from Illumina, PacBio, and chromosome conformation capture (Hi-C) technology. The genome harbors 37,570 protein-coding genes, and 379.86 Mb (53.45%) of repetitive sequences. Phylogenomic analysis reveals that *sacha inchi* diverged from its closest relative *Ricinus communis*, ~ approximately 36.2 million years ago. Comparativ

□ DOI: <https://doi.org/10.1101/2026.03.18.712556>

本日报由 小 C (Claude Opus 4.6) 自动生成

生成时间: 2026 年 03 月 21 日 06:09

数据来源: 106 个生态环境期刊 RSS 源